Model Architecture: gemma-4-E2B

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1 Model Overview

Model name	gemma-4-E2B
Transformers class	`Gemma4ForConditionalGeneration`
Path	`/models/memorization/gemma-4-E2B`
Total parameters	5,104,297,504 (5.10B)
Weight dtype	bfloat16
Hidden size d	1,536
Decoder layers L	35
Query heads H	8
Key/value heads H_{kv}	1
Head dim d_h	256
MLP width d_{ff}	6,144
PLE dim per layer	256
KV-shared layers	20
Vocabulary V	262,144
Max context length	131,072
Sliding window w	512
Logit soft-cap	30.0
RMSNorm ϵ	1e-06
Tied input/output embeddings	True
Activation	$\text{GELU}_{\text{tanh}}$
Vision encoder: layers	16
hidden size / heads	768 / 12
Audio encoder: layers	12
hidden size / heads	1024 / 8

How to read this document. Section 2 treats the model mathematically, top-down: first the whole network as one composition $\Psi \circ \mathcal{B}^{(L)} \circ \cdots \circ \mathcal{B}^{(1)} \circ E$ of layers, then a definition of every layer down to single neurons and attention heads, then parameters, outputs, intermediate states and the local linear response. Section 3 gives readable pseudocode for the forward pass and for every distinct block type. Section 4 lists the weights of each block type *once*, with how many times it repeats. Section 5 explains in plain language what every kind of layer does. The original layer-by-layer table with every repetition written out is in Appendix A.

Basic notation

B, T	batch size, sequence length (tokens)
d	hidden size (1536)
V	vocabulary size (262,144)
H, H_{kv}, d_h	query heads, key/value heads, size of one head
W_*, b_*	learned weight matrix / bias of the named projection
$\odot$, σ	element-wise product, logistic sigmoid
$f^{\circ n}$	f applied n times in a row ($f \circ f \circ \cdots \circ f$)
$g \circ f$	first f, then g (composition reads right to left)
$\text{RMSNorm}(x)$	$\gamma \odot x / \sqrt{\text{mean}(x^2) + \epsilon}$ (learned scale γ)
$\text{LN}(x)$	LayerNorm: $\gamma \odot (x - \mu)/\sigma + \beta$
RoPE	rotary position embedding (rotates query/key pairs by an angle set by position)

2 Mathematical Representation

We describe the model *top-down*. First the whole network is written as one composition of functions (Section 2.1). Then every function that appears in that composition is defined (Section 2.2), down to individual neurons and attention heads. The remaining subsections describe the parameters, the outputs, the intermediate states, and the local linear behaviour.

2.1 The model as a composition of layers

Definition 2.1 (gemma-4-E2B as an L-layer decoder)**.** The model is a function $f_\theta : \mathcal{X} \to \mathcal{Y}$ with parameters θ. It maps a sequence of T token ids $x \in \mathcal{X} = \{1, \ldots, V\}^T$ to one logit vector per position, $y \in \mathcal{Y} = \mathbb{R}^{T \times V}$

($V = 262{,}144$). It is a composition of an embedding E_{mm}, $L = 35$ decoder blocks $\mathcal{B}^{(\ell)}$ and an output head Ψ:

$$f_\theta \;=\; \Psi \circ \mathcal{B}^{(L)} \circ \mathcal{B}^{(L-1)} \circ \cdots \circ \mathcal{B}^{(2)} \circ \mathcal{B}^{(1)} \circ E_{\text{mm}} \tag{1}$$

Every block maps $\mathbb{R}^{T\times d} \to \mathbb{R}^{T\times d}$ ($d = 1536$), so the blocks can be stacked in any number. Block ℓ has its own parameters θ_ℓ and is stored in the checkpoint as `model.language_model.layers.`$(\ell - 1)$. Each block also receives its own per-layer embedding p_ℓ (Definition 2.5).

Definition 2.2 (Block types)**.** There are 4 block types (same function form within a type, different weights). A type map $c : \{1, \ldots, L\} \to \{\text{S1}, \text{F1}, \text{S2}, \text{F2}\}$ says which type layer ℓ is: $\mathcal{B}^{(\ell)} = \mathcal{B}_{c(\ell)}(\,\cdot\,; \theta_\ell)$. Here $c = ((\text{S1}^4\,\text{F1})^3\,(\text{S2}^4\,\text{F2})^4)$, read left to right from layer 1. Written out, with the type as a subscript:

$$f_\theta = \Psi \circ \mathcal{B}^{(35)}_{\text{F2}} \circ \cdots \circ \mathcal{B}^{(5)}_{\text{F1}} \circ \mathcal{B}^{(4)}_{\text{S1}} \circ \mathcal{B}^{(3)}_{\text{S1}} \circ \mathcal{B}^{(2)}_{\text{S1}} \circ \mathcal{B}^{(1)}_{\text{S1}} \circ E_{\text{mm}} \tag{2}$$

Grouping the repetitions with composition powers ($\mathcal{B}^{\circ n} = n$ consecutive blocks of the same type, each with its own weights) gives the compact form

$$f_\theta = \Psi \circ \Big(\big(\mathcal{B}_{\text{F2}} \circ \mathcal{B}_{\text{S2}}^{\circ 4}\big)^{\circ 4} \circ \big(\mathcal{B}_{\text{F1}} \circ \mathcal{B}_{\text{S1}}^{\circ 4}\big)^{\circ 3} \Big) \circ E_{\text{mm}} \tag{3}$$

Definition 2.3 (Vision encoder)**.** The vision encoder is itself a composition of 16 blocks acting on a sequence of image patches of width 768:

$$\text{Enc}_{\text{V}} = \text{Post} \circ \mathcal{B}_{\text{V}}^{\circ 16} \circ \text{PatchEmbed} \tag{4}$$

PatchEmbed turns the raw input into vectors and Post is the final norm / pooling / merging (Table 3).

Definition 2.4 (Audio encoder)**.** The audio encoder is itself a composition of 12 blocks acting on a sequence of audio frames of width 1024:

$$\text{Enc}_{\text{A}} = \text{Post} \circ \mathcal{B}_{\text{A}}^{\circ 12} \circ \text{Subsample} \tag{5}$$

Subsample turns the raw input into vectors and Post is the final norm / pooling / merging (Table 3).

Layer pattern.

Symbol	Block type	Count	Params / block	Checkpoint indices
$\mathcal{B}_{\text{S1}}$	Sliding-window attention block (MLP width 6144)	12	36,184,064	0–3, 5–8, 10–13
$\mathcal{B}_{\text{F1}}$	Full (global) attention block (MLP width 6144)	3	43,262,464	4, 9, …, 14 (every 5th)
$\mathcal{B}_{\text{S2}}$	Sliding-window attention block (KV-shared, MLP width 12288)	16	63,708,928	15–18, 20–23, 25–28, 30–33
$\mathcal{B}_{\text{F2}}$	Full (global) attention block (KV-shared, MLP width 12288)	4	70,000,640	19, 24, …, 34 (every 5th)
$\mathcal{B}_{\text{V}}$	Vision encoder block	16	9,440,384	0–15
$\mathcal{B}_{\text{A}}$	Conformer audio block	12	25,180,288	0–11

Language-model decoder (35 layers). layers 0–14: $[\mathcal{B}_{\text{S1}}\times 4, \mathcal{B}_{\text{F1}}]$ repeated 3×; layers 15–34: $[\mathcal{B}_{\text{S2}}\times 4, \mathcal{B}_{\text{F2}}]$ repeated 4× (checkpoint indices, i.e. $\ell - 1$).

Vision encoder (16 layers). layers 0–15: $[\mathcal{B}_{\text{V}}]$ repeated 16× (checkpoint indices, i.e. $\ell - 1$).

Audio encoder (12 layers). layers 0–11: $[\mathcal{B}_{\text{A}}]$ repeated 12× (checkpoint indices, i.e. $\ell - 1$).

2.2 The layers, top to bottom

Every function in the composition above is defined here. Only the attention (or linear-attention) sub-layer mixes information *between* positions t. Every other sub-layer (norms, feed-forward networks, projections) acts on each token vector $h_t \in \mathbb{R}^d$ independently, e.g. $\text{FFN}(h)_t = \text{FFN}(h_t)$. All weights follow the convention $W \in \mathbb{R}^{d_{\text{out}}\times d_{\text{in}}}$, so a projection of a token vector is Wu (as in an affine layer $f_i(u) = A^{(i)}u + b^{(i)}$).

Definition 2.5 (Token embedding)**.** $E : \{1, \ldots, V\}^T \to \mathbb{R}^{T\times d}$ looks up one row of the embedding matrix $W_E \in \mathbb{R}^{V\times d}$ per token:

$$E(x)_t = W_E[x_t, :] \cdot \sqrt{d}, \qquad t = 1, \ldots, T. \tag{6}$$

With images or audio, the multimodal embedding replaces the placeholder-token rows of $E(x)$ by projected encoder outputs:

$$E_{\mathrm{mm}}(x,\cdot) = \mathrm{Merge}\Big(E(x),\ \mathrm{Proj}_{\mathrm{img}}\big(\mathrm{Enc}_{\mathrm{V}}(\mathrm{img})\big),\ \mathrm{Proj}_{\mathrm{aud}}\big(\mathrm{Enc}_{\mathrm{A}}(\mathrm{aud})\big)\Big). \tag{7}$$

Per-layer embeddings (PLE): a second table gives every token one small vector for each layer,

$$p = \tfrac{1}{\sqrt{2}}\Big(\mathrm{Embed}_{\mathrm{PLE}}(x) + \mathrm{RMSNorm}\big(W_{\mathrm{ple}}\,E(x)/\sqrt{d}\big)\Big), \qquad p = (p_1,\dots,p_L),\ p_\ell \in \mathbb{R}^{T\times 256}. \tag{8}$$

Definition 2.6 (Normalisation)**.** $\mathrm{RMSNorm}(u) = \gamma \odot u/\sqrt{\frac{1}{d}\sum_i u_i^2 + \epsilon}$, with a learned scale $\gamma \in \mathbb{R}^d$, applied to each token vector separately. It fixes the scale of a vector without changing its direction. Every norm in the model has its own γ, distinguished by a subscript (e.g. $\mathrm{RMSNorm}_1$, $\mathrm{RMSNorm}_{\mathrm{final}}$).

Definition 2.7 (Block $\mathcal{B}_{\mathrm{S1}}$: Sliding-window attention block (MLP width 6144))**.** A decoder block of this type maps $h \in \mathbb{R}^{T\times 1536}$ to the same space. It is a composition of 3 residual sub-layers, each of the form $(\mathrm{id}+g)(h) = h + g(h)$, scaled by the fixed per-layer scalar γ_ℓ:

$$\mathcal{B}_{\mathrm{S1}} = \gamma_\ell \cdot \big((\mathrm{id}+\mathrm{PLE}_\ell)\circ(\mathrm{id}+\mathrm{RMSNorm}_{\mathrm{pf}}\circ\mathrm{FFN}\circ\mathrm{RMSNorm}_{\mathrm{pre}})\circ(\mathrm{id}+\mathrm{RMSNorm}_{\mathrm{pa}}\circ\mathrm{Attn}\circ\mathrm{RMSNorm}_{\mathrm{in}})\big) \tag{9}$$

Step by step, the sub-layers are:

$$\begin{aligned}
a &= h + \mathrm{RMSNorm}_{\mathrm{pa}}\big(\mathrm{Attn}(\mathrm{RMSNorm}_{\mathrm{in}}(h))\big) && \text{norm before AND after attention}\\
b &= a + \mathrm{RMSNorm}_{\mathrm{pf}}\big(\mathrm{FFN}(\mathrm{RMSNorm}_{\mathrm{pre}}(a))\big) && \text{norm before AND after the MLP}\\
c &= b + \mathrm{PLE}_\ell(b) && \text{inject this layer's 256-dim per-layer embedding } p_\ell\\
\mathrm{PLE}_\ell(u) &= \mathrm{RMSNorm}_{\mathrm{ple}}\big(W^{\mathrm{ple}}_{\mathrm{proj}}(\mathrm{GELU}_{\tanh}(W^{\mathrm{ple}}_{\mathrm{gate}}\,u)\odot p_\ell)\big) &&\\
\mathcal{B}(h) &= \gamma_\ell \cdot c && \text{fixed per-layer scalar}\\
Q &= W_Q\,u && H = 8 \text{ heads of size } d_h = 256\\
K,\ V &= W_K\,u,\quad W_V\,u && H_{kv} = 1 \text{ key/value heads}\\
Q,\ K &= \mathrm{RMSNorm}_q(Q),\ \mathrm{RMSNorm}_k(K) && \text{per-head normalisation}\\
V &= \mathrm{RMSNorm}(V) && \text{no learned scale}\\
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position}\\
S_h &= s\,Q_h K_{g(h)}^{\top} + M && \text{mask } M\text{: causal + sliding window of } w = 512\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\,V_{g(h)} &&\\
\mathrm{Attn}(u) &= W_O\,[O_1;\dots;O_H] &&\\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\mathrm{GELU}_{\tanh}(W_{\mathrm{gate}}\,u)\odot W_{\mathrm{up}}\,u\big) && \text{gated MLP, width 6144}
\end{aligned}$$

Parameters of one block ($|\theta_\ell| = 36{,}184{,}064$): $W_Q \in \mathbb{R}^{2048\times 1536}$, $W_K \in \mathbb{R}^{256\times 1536}$, $W_V \in \mathbb{R}^{256\times 1536}$, $W_O \in \mathbb{R}^{1536\times 2048}$, $W_{\mathrm{gate}} \in \mathbb{R}^{6144\times 1536}$, $W_{\mathrm{up}} \in \mathbb{R}^{6144\times 1536}$, $W_{\mathrm{down}} \in \mathbb{R}^{1536\times 6144}$, $W^{\mathrm{ple}}_{\mathrm{gate}} \in \mathbb{R}^{256\times 1536}$, $W^{\mathrm{ple}}_{\mathrm{proj}} \in \mathbb{R}^{1536\times 256}$, plus 8,192 parameters in norm scales, biases and other vectors. This type is used 12 times.

Definition 2.8 (Block $\mathcal{B}_{\mathrm{F1}}$: Full (global) attention block (MLP width 6144))**.** A decoder block of this type maps $h \in \mathbb{R}^{T\times 1536}$ to the same space. It is a composition of 3 residual sub-layers, each of the form $(\mathrm{id}+g)(h) = h+g(h)$, scaled by the fixed per-layer scalar γ_ℓ:

$$\mathcal{B}_{\mathrm{F1}} = \gamma_\ell \cdot \big((\mathrm{id}+\mathrm{PLE}_\ell)\circ(\mathrm{id}+\mathrm{RMSNorm}_{\mathrm{pf}}\circ\mathrm{FFN}\circ\mathrm{RMSNorm}_{\mathrm{pre}})\circ(\mathrm{id}+\mathrm{RMSNorm}_{\mathrm{pa}}\circ\mathrm{Attn}\circ\mathrm{RMSNorm}_{\mathrm{in}})\big) \tag{10}$$

Step by step, the sub-layers are:

$$\begin{aligned}
a &= h + \mathrm{RMSNorm}_{\mathrm{pa}}\big(\mathrm{Attn}(\mathrm{RMSNorm}_{\mathrm{in}}(h))\big) && \text{norm before AND after attention}\\
b &= a + \mathrm{RMSNorm}_{\mathrm{pf}}\big(\mathrm{FFN}(\mathrm{RMSNorm}_{\mathrm{pre}}(a))\big) && \text{norm before AND after the MLP}\\
c &= b + \mathrm{PLE}_\ell(b) && \text{inject this layer's 256-dim per-layer embedding } p_\ell\\
\mathrm{PLE}_\ell(u) &= \mathrm{RMSNorm}_{\mathrm{ple}}\big(W^{\mathrm{ple}}_{\mathrm{proj}}(\mathrm{GELU}_{\tanh}(W^{\mathrm{ple}}_{\mathrm{gate}}\,u)\odot p_\ell)\big) &&\\
\mathcal{B}(h) &= \gamma_\ell \cdot c && \text{fixed per-layer scalar}\\
Q &= W_Q\,u && H = 8 \text{ heads of size } d_h = 512\\
K,\ V &= W_K\,u,\quad W_V\,u && H_{kv} = 1 \text{ key/value heads}\\
Q,\ K &= \mathrm{RMSNorm}_q(Q),\ \mathrm{RMSNorm}_k(K) && \text{per-head normalisation}
\end{aligned}$$

$$
\begin{aligned}
V &= \mathrm{RMSNorm}(V) && \text{no learned scale}\\
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position}\\
S_h &= s\, Q_h K_{g(h)}^{\top} + M && \text{mask } M\text{: causal}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\, V_{g(h)}\\
\mathrm{Attn}(u) &= W_O\,[O_1;\ldots;O_H]\\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\,\mathrm{GELU}_{\tanh}(W_{\mathrm{gate}}\,u)\odot W_{\mathrm{up}}\,u\big) && \text{gated MLP, width 6144}
\end{aligned}
$$

Parameters of one block ($|\theta_\ell| = 43{,}262{,}464$): $W_Q \in \mathbb{R}^{4096\times 1536}$, $W_K \in \mathbb{R}^{512\times 1536}$, $W_V \in \mathbb{R}^{512\times 1536}$, $W_O \in \mathbb{R}^{1536\times 4096}$, $W_{\mathrm{gate}} \in \mathbb{R}^{6144\times 1536}$, $W_{\mathrm{up}} \in \mathbb{R}^{6144\times 1536}$, $W_{\mathrm{down}} \in \mathbb{R}^{1536\times 6144}$, $W^{\mathrm{ple}}_{\mathrm{gate}} \in \mathbb{R}^{256\times 1536}$, $W^{\mathrm{ple}}_{\mathrm{proj}} \in \mathbb{R}^{1536\times 256}$, plus 8,704 parameters in norm scales, biases and other vectors. This type is used 3 times.

Definition 2.9 (Block $\mathcal{B}_{\mathrm{S2}}$: Sliding-window attention block (KV-shared, MLP width 12288))**.** A decoder block of this type maps $h \in \mathbb{R}^{T\times 1536}$ to the same space. It is a composition of 3 residual sub-layers, each of the form $(\mathrm{id}+g)(h) = h + g(h)$, scaled by the fixed per-layer scalar γ_ℓ:

$$
\mathcal{B}_{\mathrm{S2}} = \gamma_\ell \cdot \big((\mathrm{id}+\mathrm{PLE}_\ell)\circ(\mathrm{id}+\mathrm{RMSNorm}_{\mathrm{pf}}\circ\mathrm{FFN}\circ\mathrm{RMSNorm}_{\mathrm{pre}})\circ(\mathrm{id}+\mathrm{RMSNorm}_{\mathrm{pa}}\circ\mathrm{Attn}\circ\mathrm{RMSNorm}_{\mathrm{in}})\big) \quad (11)
$$

Step by step, the sub-layers are:

$$
\begin{aligned}
a &= h + \mathrm{RMSNorm}_{\mathrm{pa}}\big(\,\mathrm{Attn}(\mathrm{RMSNorm}_{\mathrm{in}}(h))\big) && \text{norm before AND after attention}\\
b &= a + \mathrm{RMSNorm}_{\mathrm{pf}}\big(\,\mathrm{FFN}(\mathrm{RMSNorm}_{\mathrm{pre}}(a))\big) && \text{norm before AND after the MLP}\\
c &= b + \mathrm{PLE}_\ell(b) && \text{inject this layer's 256-dim per-layer embedding } p_\ell\\
\mathrm{PLE}_\ell(u) &= \mathrm{RMSNorm}_{\mathrm{ple}}\big(W^{\mathrm{ple}}_{\mathrm{proj}}(\mathrm{GELU}_{\tanh}(W^{\mathrm{ple}}_{\mathrm{gate}}\,u)\odot p_\ell)\big)\\
\mathcal{B}(h) &= \gamma_\ell \cdot c && \text{fixed per-layer scalar}\\
Q &= W_Q\,u && H = 8 \text{ heads of size } d_h = 256\\
K,\,V &= K^{(\ell^*)},\,V^{(\ell^*)} && \text{from last non-shared layer } \ell^* \text{ (no K/V weights)}\\
Q,\ K &= \mathrm{RMSNorm}_q(Q),\ K && \text{per-head normalisation}\\
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position}\\
S_h &= s\, Q_h K_{g(h)}^{\top} + M && \text{mask } M\text{: causal + sliding window of } w = 512\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\, V_{g(h)}\\
\mathrm{Attn}(u) &= W_O\,[O_1;\ldots;O_H]\\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\,\mathrm{GELU}_{\tanh}(W_{\mathrm{gate}}\,u)\odot W_{\mathrm{up}}\,u\big) && \text{gated MLP, width 12288}
\end{aligned}
$$

Parameters of one block ($|\theta_\ell| = 63{,}708{,}928$): $W_Q \in \mathbb{R}^{2048\times 1536}$, $W_O \in \mathbb{R}^{1536\times 2048}$, $W_{\mathrm{gate}} \in \mathbb{R}^{12288\times 1536}$, $W_{\mathrm{up}} \in \mathbb{R}^{12288\times 1536}$, $W_{\mathrm{down}} \in \mathbb{R}^{1536\times 12288}$, $W^{\mathrm{ple}}_{\mathrm{gate}} \in \mathbb{R}^{256\times 1536}$, $W^{\mathrm{ple}}_{\mathrm{proj}} \in \mathbb{R}^{1536\times 256}$, plus 7,936 parameters in norm scales, biases and other vectors. This type is used 16 times.

Definition 2.10 (Block $\mathcal{B}_{\mathrm{F2}}$: Full (global) attention block (KV-shared, MLP width 12288))**.** A decoder block of this type maps $h \in \mathbb{R}^{T\times 1536}$ to the same space. It is a composition of 3 residual sub-layers, each of the form $(\mathrm{id}+g)(h) = h + g(h)$, scaled by the fixed per-layer scalar γ_ℓ:

$$
\mathcal{B}_{\mathrm{F2}} = \gamma_\ell \cdot \big((\mathrm{id}+\mathrm{PLE}_\ell)\circ(\mathrm{id}+\mathrm{RMSNorm}_{\mathrm{pf}}\circ\mathrm{FFN}\circ\mathrm{RMSNorm}_{\mathrm{pre}})\circ(\mathrm{id}+\mathrm{RMSNorm}_{\mathrm{pa}}\circ\mathrm{Attn}\circ\mathrm{RMSNorm}_{\mathrm{in}})\big) \quad (12)
$$

Step by step, the sub-layers are:

$$
\begin{aligned}
a &= h + \mathrm{RMSNorm}_{\mathrm{pa}}\big(\,\mathrm{Attn}(\mathrm{RMSNorm}_{\mathrm{in}}(h))\big) && \text{norm before AND after attention}\\
b &= a + \mathrm{RMSNorm}_{\mathrm{pf}}\big(\,\mathrm{FFN}(\mathrm{RMSNorm}_{\mathrm{pre}}(a))\big) && \text{norm before AND after the MLP}\\
c &= b + \mathrm{PLE}_\ell(b) && \text{inject this layer's 256-dim per-layer embedding } p_\ell\\
\mathrm{PLE}_\ell(u) &= \mathrm{RMSNorm}_{\mathrm{ple}}\big(W^{\mathrm{ple}}_{\mathrm{proj}}(\mathrm{GELU}_{\tanh}(W^{\mathrm{ple}}_{\mathrm{gate}}\,u)\odot p_\ell)\big)\\
\mathcal{B}(h) &= \gamma_\ell \cdot c && \text{fixed per-layer scalar}\\
Q &= W_Q\,u && H = 8 \text{ heads of size } d_h = 512\\
K,\,V &= K^{(\ell^*)},\,V^{(\ell^*)} && \text{from last non-shared layer } \ell^* \text{ (no K/V weights)}\\
Q,\ K &= \mathrm{RMSNorm}_q(Q),\ K && \text{per-head normalisation}\\
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position}
\end{aligned}
$$

$$
\begin{aligned}
S_h &= s\, Q_h K_{g(h)}^\top + M && \text{mask } M\text{: causal}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\, V_{g(h)} \\
\mathrm{Attn}(u) &= W_O\,[O_1;\ldots;O_H] \\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\,\mathrm{GELU}_{\tanh}(W_{\mathrm{gate}}\,u)\odot W_{\mathrm{up}}\,u\big) && \text{gated MLP, width 12288}
\end{aligned}
$$

Parameters of one block ($|\theta_\ell| = 70{,}000{,}640$): $W_Q \in \mathbb{R}^{4096\times1536}$, $W_O \in \mathbb{R}^{1536\times4096}$, $W_{\mathrm{gate}} \in \mathbb{R}^{12288\times1536}$, $W_{\mathrm{up}} \in \mathbb{R}^{12288\times1536}$, $W_{\mathrm{down}} \in \mathbb{R}^{1536\times12288}$, $W^{\mathrm{ple}}_{\mathrm{gate}} \in \mathbb{R}^{256\times1536}$, $W^{\mathrm{ple}}_{\mathrm{proj}} \in \mathbb{R}^{1536\times256}$, plus 8,192 parameters in norm scales, biases and other vectors. This type is used 4 times.

Definition 2.11 (Block $\mathcal{B}_{\mathrm{V}}$: Vision encoder block)**.** A vision-encoder block of this type maps $h \in \mathbb{R}^{T\times768}$ to the same space. It is a composition of 2 residual sub-layers, each of the form $(\mathrm{id}+g)(h) = h + g(h)$:

$$
\mathcal{B}_{\mathrm{V}} = (\mathrm{id} + \mathrm{RMSNorm}_{\mathrm{pf}} \circ \mathrm{FFN} \circ \mathrm{RMSNorm}_{\mathrm{pre}}) \circ (\mathrm{id} + \mathrm{RMSNorm}_{\mathrm{pa}} \circ \mathrm{Attn} \circ \mathrm{RMSNorm}_{\mathrm{in}}) \tag{13}
$$

Step by step, the sub-layers are:

$$
\begin{aligned}
a &= h + \mathrm{RMSNorm}_{\mathrm{pa}}\big(\,\mathrm{Attn}(\mathrm{RMSNorm}_{\mathrm{in}}(h))\big) && \text{norm before AND after attention}\\
\mathcal{B}(h) &= a + \mathrm{RMSNorm}_{\mathrm{pf}}\big(\,\mathrm{FFN}(\mathrm{RMSNorm}_{\mathrm{pre}}(a))\big) && \text{norm before AND after the MLP}\\
Q &= W_Q\, u && H = 12 \text{ heads of size } d_h = 64\\
K,\ V &= W_K\, u, \quad W_V\, u && H_{kv} = 12 \text{ key/value heads}\\
Q,\ K &= \mathrm{RMSNorm}_q(Q),\ \mathrm{RMSNorm}_k(K) && \text{per-head normalisation}\\
V &= \mathrm{RMSNorm}(V) && \text{no learned scale}\\
Q,\ K &= \mathrm{RoPE}_{2D}(Q),\ \mathrm{RoPE}_{2D}(K) && \text{rotate by (row, column) patch position}\\
S_h &= s\, Q_h K_h^\top + M && \text{mask } M\text{: bidirectional}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\, V_h \\
\mathrm{Attn}(u) &= W_O\,[O_1;\ldots;O_H] \\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\,\mathrm{GELU}_{\tanh}(W_{\mathrm{gate}}\,u)\odot W_{\mathrm{up}}\,u\big) && \text{gated MLP, width 3072}
\end{aligned}
$$

Parameters of one block ($|\theta_\ell| = 9{,}440{,}384$): $W_Q \in \mathbb{R}^{768\times768}$, $W_K \in \mathbb{R}^{768\times768}$, $W_V \in \mathbb{R}^{768\times768}$, $W_O \in \mathbb{R}^{768\times768}$, $W_{\mathrm{gate}} \in \mathbb{R}^{3072\times768}$, $W_{\mathrm{up}} \in \mathbb{R}^{3072\times768}$, $W_{\mathrm{down}} \in \mathbb{R}^{768\times3072}$, plus 3,200 parameters in norm scales, biases and other vectors. This type is used 16 times.

Definition 2.12 (Block $\mathcal{B}_{\mathrm{A}}$: Conformer audio block)**.** A audio-encoder block of this type maps $h \in \mathbb{R}^{T\times1024}$ to the same space. It is a composition of 4 residual sub-layers, each of the form $(\mathrm{id}+g)(h) = h + g(h)$, followed by a final norm:

$$
\mathcal{B}_{\mathrm{A}} = \mathrm{RMSNorm}_{\mathrm{out}} \circ (\mathrm{id} + \tfrac{1}{2}\,\mathrm{FFN}_2) \circ (\mathrm{id} + \mathrm{Conv}) \circ (\mathrm{id} + \mathrm{RMSNorm}_{\mathrm{post}} \circ \mathrm{Attn}_{rel} \circ \mathrm{RMSNorm}_{\mathrm{pre}}) \circ (\mathrm{id} + \tfrac{1}{2}\,\mathrm{FFN}_1) \tag{14}
$$

Step by step, the sub-layers are:

$$
\begin{aligned}
h_1 &= h + \tfrac{1}{2}\ \mathrm{FFN}_1(h) && \text{macaron half-step FFN}\\
h_2 &= h_1 + \mathrm{RMSNorm}_{\mathrm{post}}\big(\,\mathrm{Attn}_{rel}(\mathrm{RMSNorm}_{\mathrm{pre}}(h_1))\big) && \text{local attention, relative positions}\\
h_3 &= h_2 + W_{\mathrm{end}}\ \mathrm{SiLU}\big(\,\mathrm{RMSNorm}(\mathrm{DWConv}(\mathrm{GLU}(W_{\mathrm{start}}\,\mathrm{RMSNorm}(h_2))))\big) && \text{convolution module}\\
h_4 &= h_3 + \tfrac{1}{2}\ \mathrm{FFN}_2(h_3) \\
\mathcal{B}(h) &= \mathrm{RMSNorm}_{\mathrm{out}}(h_4) \\
\mathrm{FFN}_i(u) &= \mathrm{RMSNorm}\,\big(W_2\ \mathrm{SiLU}(W_1\ \mathrm{RMSNorm}(u))\big)
\end{aligned}
$$

Parameters of one block ($|\theta_\ell| = 25{,}180{,}288$): $W_1^{[1]} \in \mathbb{R}^{4096\times1024}$, $W_2^{[1]} \in \mathbb{R}^{1024\times4096}$, $W_1^{[2]} \in \mathbb{R}^{4096\times1024}$, $W_2^{[2]} \in \mathbb{R}^{1024\times4096}$, $W_Q \in \mathbb{R}^{1024\times1024}$, $W_K \in \mathbb{R}^{1024\times1024}$, $W_V \in \mathbb{R}^{1024\times1024}$, $W_O \in \mathbb{R}^{1024\times1024}$, $W_{\mathrm{rel}} \in \mathbb{R}^{1024\times1024}$, $W_{\mathrm{start}} \in \mathbb{R}^{2048\times1024}$, $W_{\mathrm{end}} \in \mathbb{R}^{1024\times1024}$, $K_{\mathrm{dw}} \in \mathbb{R}^{1024\times1\times5}$, plus 9,344 parameters in norm scales, biases and other vectors. This type is used 12 times.

Definition 2.13 (Output head)**.** $\Psi : \mathbb{R}^{T\times d} \to \mathbb{R}^{T\times V}$ normalises the last hidden state and scores every vocabulary token, re-using the embedding matrix (tied weights):

$$
\Psi(h)_t = 30.0 \cdot \tanh\big(W_E^\top\ \mathrm{RMSNorm}_{\mathrm{final}}(h)_t / 30.0\big) \tag{15}
$$

The tanh soft-cap keeps every logit in $(-30.0, 30.0)$.

2.3 Neurons and attention heads

Definition 2.14 (Neuron, pre-activation value and state)**.** The feed-forward sub-layer of block ℓ contains $d_{ff}^{(\ell)} \in \{6144, 12288\}$ neurons. Neuron j of layer ℓ is the scalar function of a normalised token vector $u \in \mathbb{R}^d$

$$\eta_j^{(\ell)}(u) = \mathrm{GELU}_{\tanh}\left(W_{\mathrm{gate},j}^{(\ell)} \cdot u\right)\left(W_{\mathrm{up},j}^{(\ell)} \cdot u\right), \tag{16}$$

where $W_{\mathrm{gate},j}$ is the j-th row of W_{gate}. The feed-forward output is the sum of the neuron outputs, each written along its own column of W_{down}:

$$\mathrm{FFN}_\ell(u) = \sum_{j=1}^{d_{ff}^{(\ell)}} \eta_j^{(\ell)}(u)\, W_{\mathrm{down}}^{(\ell)}[:, j]. \tag{17}$$

Across its L blocks the network has $N = \sum_{\ell=1}^{L} d_{ff}^{(\ell)} = 337{,}920$ feed-forward neurons. For input x at position t, with $u_t^{(\ell)}(x)$ the normalised input of the feed-forward sub-layer:

- $\mathcal{V}(\eta; x, t) = W_{\mathrm{gate},j}^{(\ell)} \cdot u_t^{(\ell)}(x) \in \mathbb{R}$ is the **pre-activation value** of neuron η (the argument of the non-linearity).
- $\mathcal{S}(\eta; x, t)$ is its **activation state**: *active* if $\mathcal{V} > 0$ and *inactive* if $\mathcal{V} \le 0$. The non-linearity is smooth, unlike ReLU, so an inactive neuron's output is strongly suppressed rather than exactly zero.

Definition 2.15 (Attention head and attention pattern)**.** Head h of an attention layer ℓ reads from the whole sequence through its attention pattern $P_h^{(\ell)}(x) \in \mathbb{R}^{T \times T}$ (row t = how much position t attends to each earlier position) and writes through its own $d \times d_h$ slice $W_{O,h}^{(\ell)}$ of W_O:

$$\mathrm{Attn}_\ell(u) = \sum_{h=1}^{H} P_h^{(\ell)}\, V_{g(h)}\, W_{O,h}^{(\ell)\top}. \tag{18}$$

$g(h)$ is the key/value head shared by query head h (grouped-query attention). The pattern P_h plays the role that the activation state plays for neurons: once P_h is fixed, the head is linear in its values. The network has 35 attention layers $\times$ $H = 8$ heads $= 280$ attention heads.

2.4 Architecture and parameters

Definition 2.16 (Architecture and parameters)**.** The **architecture** is the tuple of sizes and the layer-type sequence:

$$\mathcal{A} = (V, d, L, c, H, H_{kv}, d_h, d_{ff}) = (262{,}144,\ 1536,\ 35,\ ((\mathrm{S1}^4\,\mathrm{F1})^3\,(\mathrm{S2}^4\,\mathrm{F2})^4),\ 8,\ 1,\ \{256, 512\},\ \{6144, 12288\}). \tag{19}$$

The **parameters** are $\theta = \{W_E,\ \theta_1, \ldots, \theta_L,\ \gamma_{\mathrm{final}},\ \theta_{\mathrm{Enc_V}},\ \theta_{\mathrm{Enc_A}},\ \theta_{\mathrm{Proj}},\ \theta_{\mathrm{PLE}}\}$ (W_E is shared with the output head). Grouping the blocks by type,

$$\begin{aligned}
|\theta| &= \sum_{\text{types } c} n_c\, |\theta_c| + |\theta_{\mathrm{rest}}| \\
&= 12 \times 36{,}184{,}064 + 3 \times 43{,}262{,}464 + 16 \times 63{,}708{,}928 \\
&\quad + 4 \times 70{,}000{,}640 + 16 \times 9{,}440{,}384 + 12 \times 25{,}180{,}288 \\
&\quad + 2{,}787{,}746{,}336 \\
&= 5{,}104{,}297{,}504,
\end{aligned}$$

where n_c is how often type c occurs and θ_{rest} holds everything outside the repeated stacks (embedding, final norm, encoder stems and projector, …; Table 3).

2.5 Output representations

Definition 2.17 (Output representations)**.** For an input sequence x, the output can be read at three levels:

1. **Logits:** $y = f_\theta(x) \in \mathbb{R}^{T \times V}$, one score per vocabulary token at every position.
2. **Next-token distribution:** $p(x)_{t,k} = \dfrac{\exp(y_{t,k})}{\sum_{j=1}^{V} \exp(y_{t,j})} = \Pr\left(x_{t+1} = k \mid x_1, \ldots, x_t\right)$.
3. **Predicted next token:** $z(x) = \mathrm{argmax}_{k \in \{1,\ldots,V\}}\, y_{T,k}$ (greedy decoding). Generation appends $z(x)$ (or a sample from $p(x)_T$) to x and applies f_θ again; in practice the keys and values of earlier positions are cached.

The model is **causal**: y_t depends only on $x_1, \dots, x_t$, because every attention mask hides later positions (sliding-window layers additionally hide positions more than w tokens back). Training minimises the cross-entropy $-\sum_t \log p(x)_{t,x_{t+1}}$.

2.6 Intermediate states and sub-network decomposition

To study what happens at one layer $\ell \in \{1, \dots, L\}$, split the composition into a prefix, the layer itself and a suffix:

$$f_\theta = \underbrace{\Psi \circ \mathcal{B}^{(L)} \circ \cdots \circ \mathcal{B}^{(\ell+1)}}_{G_{\ell+1}\ \text{(suffix)}} \circ \underbrace{\mathcal{B}^{(\ell)}}_{\text{layer } \ell} \circ \underbrace{\mathcal{B}^{(\ell-1)} \circ \cdots \circ \mathcal{B}^{(1)} \circ E_{\text{mm}}}_{F_{\ell-1}\ \text{(prefix)}} \tag{20}$$

- **Prefix** $F_{\ell-1} : \mathcal{X} \to \mathbb{R}^{T \times d}$ maps the tokens to the hidden state entering layer ℓ ($F_0 = E_{\text{mm}}$).
- **Hidden state** (residual stream) after layer ℓ: $h^{(\ell)} = F_\ell(x) = \mathcal{B}^{(\ell)}(h^{(\ell-1)}, p_\ell) \in \mathbb{R}^{T \times d}$, with $h^{(0)} = E_{\text{mm}}(x)$.
- **Inside layer** ℓ**:** the attention write $a^{(\ell)} = \text{RMSNorm}_{\text{pa}}\big(\text{Attn}_\ell(\text{RMSNorm}_{\text{in}}(h^{(\ell-1)}))\big)$, the mid-block state $\hat{h}^{(\ell)} = h^{(\ell-1)} + a^{(\ell)}$, the feed-forward input $u^{(\ell)} = \text{RMSNorm}_{\text{pre}}(\hat{h}^{(\ell)})$, the pre-activations $\hat{x}^{(\ell)} = u^{(\ell)} W_{\text{gate}}^{(\ell)\top} \in \mathbb{R}^{T \times d_{ff}}$ and the neuron outputs $\eta^{(\ell)}(u^{(\ell)}) \in \mathbb{R}^{T \times d_{ff}}$, and the feed-forward write $m^{(\ell)} = \text{RMSNorm}_{\text{pf}}\big(\text{FFN}_\ell(u^{(\ell)})\big)$.
- **Suffix** $G_{\ell+1} : \mathbb{R}^{T \times d} \to \mathbb{R}^{T \times V}$ maps $h^{(\ell)}$ to the logits ($G_{L+1} = \Psi$).

Residual stream. Because every sub-layer only *adds* to the hidden state, unrolling the composition gives the final hidden state as the embedding plus everything the layers wrote:

$$h^{(L)} = \Big(\prod_{k=1}^{L} \gamma_k\Big)\, h^{(0)} + \sum_{\ell=1}^{L} \Big(\prod_{k=\ell}^{L} \gamma_k\Big)\big(a^{(\ell)} + m^{(\ell)} + c^{(\ell)}\big) \tag{21}$$

with $c^{(\ell)} = \text{PLE}_\ell(\hat{h}^{(\ell)} + m^{(\ell)})$ the per-layer-embedding write. The fixed scalars γ_ℓ rescale everything written so far, so earlier writes are multiplied by all later scalars. So $f_\theta(x) = \Psi(h^{(L)})$: each layer reads the current stream and adds a correction to it.

2.7 Local linear representation

A ReLU network is piecewise affine: inside a polytope cell it collapses to one matrix. This model is not, because softmax, $\text{GELU}_{\text{tanh}}$ and the normalisations are smooth. Its local behaviour is described by the first-order (Jacobian) linearisation instead. Tokens are discrete, so we perturb the embedded input $h^{(0)}$.

Definition 2.18 (Local linear response)**.** Let $J_\ell = D\mathcal{B}^{(\ell)}(h^{(\ell-1)})$ be the Jacobian of block ℓ at the current hidden state (tensors flattened to vectors). The chain rule applied to the residual sub-layers $(\text{id} + g_{\ell,i})$ gives

$$J_\ell = \gamma_\ell\,(I + Dg_{\ell,3})(I + Dg_{\ell,2})(I + Dg_{\ell,1}), \tag{22}$$

where $g_{\ell,1}$ is the attention branch, $g_{\ell,2}$ the feed-forward branch, $g_{\ell,3}$ the PLE branch and each Dg is taken at that branch's input. For a small perturbation δ of $h^{(0)}$, the response of the network decomposed around layer ℓ is

$$f_\theta(h^{(0)} + \delta) - f_\theta(h^{(0)}) \approx G_x^{(\ell+1)} \cdot J_\ell \cdot F_x^{(\ell-1)} \cdot \delta, \tag{23}$$

with $F_x^{(\ell-1)} = J_{\ell-1} \cdots J_1$ (prefix) and $G_x^{(\ell+1)} = J_\Psi\, J_L \cdots J_{\ell+1}$ (suffix). Multiplying out the products of $(I + Dg)$ terms writes $J_\Psi J_L \cdots J_1$ as a sum over *paths*: the identity path straight from embedding to head, plus one term for every subset of sub-layers a signal passes through.

2.8 Summary of notation

Symbol	Meaning
$f_\theta : \mathcal{X} \to \mathcal{Y}$	the whole model, $\{1..V\}^T \to \mathbb{R}^{T \times V}$
$E_{\text{mm}},\ \Psi$	embedding and output head
$\text{Enc}_{\text{V}},\ \text{Enc}_{\text{A}},\ \text{Proj}$	vision / audio encoders and their projectors into the text width
$\mathcal{B}^{(\ell)},\ \theta_\ell$	block ℓ and its parameters, $\ell = 1, \dots, L$
p_ℓ	per-layer embedding fed to block ℓ

continued on next page

Symbol	Meaning
$\mathcal{B}_c,\ c(\ell)$	block type c; type of layer ℓ
$\mathcal{B}^{\circ n}$	n consecutive blocks of one type (each with its own weights)
$(\mathrm{id} + g)$	residual sub-layer $h \mapsto h + g(h)$
$h^{(\ell)}$	hidden state (residual stream) after layer ℓ; $h^{(0)}$ = embeddings
$a^{(\ell)},\ m^{(\ell)}$	what the attention / feed-forward sub-layer of layer ℓ adds to the stream
$\eta_j^{(\ell)}$	j-th feed-forward neuron of layer ℓ
$\mathcal{V}(\eta; x, t),\ \mathcal{S}(\eta; x, t)$	pre-activation value and active/inactive state of a neuron
$P_h^{(\ell)}$	attention pattern of head h in layer ℓ ($T \times T$)
$F_{\ell-1},\ G_{\ell+1}$	prefix (embedding to layer $\ell - 1$) and suffix (layer $\ell + 1$ to logits)
$J_\ell,\ F_x^{(\ell-1)},\ G_x^{(\ell+1)}$	Jacobian of block ℓ, and of the prefix / suffix at input x
$p(x),\ z(x)$	next-token distribution and greedy next-token prediction

3 Forward Pass in Pseudocode

Algorithm 1 gemma-4-E2B: full forward pass

Require: token ids x of shape $B \times T$; optional images / audio

$h \leftarrow E(x) \cdot \sqrt{d}$ ▷ $h^{(0)}$: one vector per token, $B{\times}T{\times}1536$
if image input present **then**
  $e \leftarrow \mathrm{Enc_V}(\text{image})$ ▷ vision encoder: 16 layers, Algorithm 2
  $h[\text{image token slots}] \leftarrow \mathrm{Proj}(e)$ ▷ project into text width 1536
end if
if audio input present **then**
  $e \leftarrow \mathrm{Enc_A}(\text{audio})$ ▷ audio encoder: 12 layers, Algorithm 3
  $h[\text{audio token slots}] \leftarrow \mathrm{Proj}(e)$ ▷ project into text width 1536
end if
$p \leftarrow \mathrm{PerLayerInputs}(x, h)$ ▷ 35 small vectors per token, one for each layer
$\ell \leftarrow 0$
for $r \leftarrow 1$ **to** 3 **do** ▷ layers 1–15: pattern of 5 blocks, repeated 3 times
  for $j \leftarrow 1$ **to** 4 **do**
    $\ell \leftarrow \ell + 1;\ h \leftarrow \mathcal{B}_{\mathrm{S1}}(h, p_\ell; \theta_\ell)$ ▷ Sliding-window attention block (MLP width 6144)
  end for
  $\ell \leftarrow \ell + 1;\ h \leftarrow \mathcal{B}_{\mathrm{F1}}(h, p_\ell; \theta_\ell)$ ▷ Full (global) attention block (MLP width 6144)
end for
$\ell \leftarrow 15$
for $r \leftarrow 1$ **to** 4 **do** ▷ layers 16–35: pattern of 5 blocks, repeated 4 times
  for $j \leftarrow 1$ **to** 4 **do**
    $\ell \leftarrow \ell + 1;\ h \leftarrow \mathcal{B}_{\mathrm{S2}}(h, p_\ell; \theta_\ell)$ ▷ Sliding-window attention block (KV-shared, MLP width 12288)
  end for
  $\ell \leftarrow \ell + 1;\ h \leftarrow \mathcal{B}_{\mathrm{F2}}(h, p_\ell; \theta_\ell)$ ▷ Full (global) attention block (KV-shared, MLP width 12288)
end for
$h \leftarrow \mathrm{RMSNorm_{final}}(h)$ ▷ $h = h^{(L)}$; this and the next step are Ψ
$y \leftarrow h\, W_E^\top$ ▷ logits $B{\times}T{\times}262,144$, embedding matrix re-used
$y \leftarrow 30.0 \tanh(y/30.0)$ ▷ soft-cap logits
return y ▷ $\mathrm{softmax}(y)$ = next-token probabilities

Algorithm 2 Vision encoder

$h \leftarrow \mathrm{PatchEmbed}(\text{pixels}) + \text{position embedding}$ ▷ one vector per image patch
for $\ell \leftarrow 1$ **to** 16 **do** ▷ 16 blocks of type $\mathcal{B}_{\mathrm{V}}$
  $h \leftarrow \mathcal{B}_{\mathrm{V}}(h; \theta_\ell)$ ▷ Vision encoder block; result is $h^{(\ell)}$
end for
$h \leftarrow \mathrm{Post}(h)$ ▷ final norm / pooling / merging, see Table 3
return h

Algorithm 3 Audio encoder

$h \leftarrow$ Subsample(mel spectrogram) ▷ 2 strided convs: 4× fewer frames
for $\ell \leftarrow 1$ **to** 12 **do** ▷ 12 blocks of type $\mathcal{B}_{\mathrm{A}}$
  $h \leftarrow \mathcal{B}_{\mathrm{A}}(h;\, \theta_\ell)$ ▷ Conformer audio block; result is $h^{(\ell)}$
end for
$h \leftarrow \mathrm{Post}(h)$ ▷ final norm / pooling / merging, see Table 3
return h

Algorithm 4 Block $\mathcal{B}_{\mathrm{S1}}(h)$ — Sliding-window attention block (MLP width 6144)

Require: h of shape $B \times T \times 1536$, per-layer input p_ℓ

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top$ ▷ $B{\times}T{\times}8{\times}256$ (8 query heads)
$K \leftarrow uW_K^\top;\ V \leftarrow uW_V^\top$ ▷ $B{\times}T{\times}1{\times}256$ (1 key/value heads, each shared by 8 query heads)
$Q \leftarrow \mathrm{RMSNorm}(Q);\ K \leftarrow \mathrm{RMSNorm}(K)$ ▷ normalise each head vector (stabilises training)
$V \leftarrow \mathrm{RMSNorm}(V)$
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\, QK^\top + M)$ ▷ mask: causal + only the last 512 tokens
$O \leftarrow PV$ ▷ $B{\times}T{\times}8{\times}256$
$m \leftarrow O\, W_O^\top$ ▷ concatenate heads, project back to $B{\times}T{\times}1536$
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-attention norm, then add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{GELU}_{\tanh}(uW_{\mathrm{gate}}^\top) \odot uW_{\mathrm{up}}^\top\big) W_{\mathrm{down}}^\top$ ▷ expand to 6144, gate, shrink back to 1536
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-feed-forward norm, then add residual
per-layer embedding injection
$h \leftarrow h + \mathrm{RMSNorm}\left((\mathrm{GELU}_{\tanh}(hW_{\mathrm{gate}}^{\mathrm{ple}\top}) \odot p_\ell) W_{\mathrm{proj}}^{\mathrm{ple}\top}\right)$ ▷ gate down to 256, multiply by p_ℓ, project back up
$h \leftarrow \gamma_\ell\, h$ ▷ fixed per-layer scale
return h

Algorithm 5 Block $\mathcal{B}_{\mathrm{F1}}(h)$ — Full (global) attention block (MLP width 6144)

Require: h of shape $B \times T \times 1536$, per-layer input p_ℓ

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top$ ▷ $B{\times}T{\times}8{\times}512$ (8 query heads)
$K \leftarrow uW_K^\top;\ V \leftarrow uW_V^\top$ ▷ $B{\times}T{\times}1{\times}512$ (1 key/value heads, each shared by 8 query heads)
$Q \leftarrow \mathrm{RMSNorm}(Q);\ K \leftarrow \mathrm{RMSNorm}(K)$ ▷ normalise each head vector (stabilises training)
$V \leftarrow \mathrm{RMSNorm}(V)$
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\, QK^\top + M)$ ▷ mask: causal: a token sees only earlier tokens
$O \leftarrow PV$ ▷ $B{\times}T{\times}8{\times}512$
$m \leftarrow O\, W_O^\top$ ▷ concatenate heads, project back to $B{\times}T{\times}1536$
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-attention norm, then add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{GELU}_{\tanh}(uW_{\mathrm{gate}}^\top) \odot uW_{\mathrm{up}}^\top\big) W_{\mathrm{down}}^\top$ ▷ expand to 6144, gate, shrink back to 1536
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-feed-forward norm, then add residual
per-layer embedding injection
$h \leftarrow h + \mathrm{RMSNorm}\left((\mathrm{GELU}_{\tanh}(hW_{\mathrm{gate}}^{\mathrm{ple}\top}) \odot p_\ell) W_{\mathrm{proj}}^{\mathrm{ple}\top}\right)$ ▷ gate down to 256, multiply by p_ℓ, project back up
$h \leftarrow \gamma_\ell\, h$ ▷ fixed per-layer scale
return h

Algorithm 6 Block $\mathcal{B}_{\mathrm{S2}}(h)$ — Sliding-window attention block (KV-shared, MLP width 12288)

Require: h of shape $B \times T \times 1536$, per-layer input p_ℓ

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top$ ▷ $B{\times}T{\times}8{\times}256$ (8 query heads)
$K, V \leftarrow K^{(\ell^*)}, V^{(\ell^*)}$ ▷ re-use keys/values of an earlier layer (saves compute and KV-cache)
$Q \leftarrow \mathrm{RMSNorm}(Q)$ ▷ normalise each head vector (stabilises training)
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: causal + only the last 512 tokens
$O \leftarrow PV$ ▷ $B{\times}T{\times}8{\times}256$
$m \leftarrow O\,W_O^\top$ ▷ concatenate heads, project back to $B{\times}T{\times}1536$
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-attention norm, then add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{GELU}_{\tanh}(uW_{\mathrm{gate}}^\top) \odot uW_{\mathrm{up}}^\top\big)W_{\mathrm{down}}^\top$ ▷ expand to 12288, gate, shrink back to 1536
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-feed-forward norm, then add residual
per-layer embedding injection
$h \leftarrow h + \mathrm{RMSNorm}\left((\mathrm{GELU}_{\tanh}(hW_{\mathrm{gate}}^{\mathrm{ple}\top}) \odot p_\ell)W_{\mathrm{proj}}^{\mathrm{ple}\top}\right)$ ▷ gate down to 256, multiply by p_ℓ, project back up
$h \leftarrow \gamma_\ell\, h$ ▷ fixed per-layer scale
return h

Algorithm 7 Block $\mathcal{B}_{\mathrm{F2}}(h)$ — Full (global) attention block (KV-shared, MLP width 12288)

Require: h of shape $B \times T \times 1536$, per-layer input p_ℓ

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top$ ▷ $B{\times}T{\times}8{\times}512$ (8 query heads)
$K, V \leftarrow K^{(\ell^*)}, V^{(\ell^*)}$ ▷ re-use keys/values of an earlier layer (saves compute and KV-cache)
$Q \leftarrow \mathrm{RMSNorm}(Q)$ ▷ normalise each head vector (stabilises training)
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: causal: a token sees only earlier tokens
$O \leftarrow PV$ ▷ $B{\times}T{\times}8{\times}512$
$m \leftarrow O\,W_O^\top$ ▷ concatenate heads, project back to $B{\times}T{\times}1536$
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-attention norm, then add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{GELU}_{\tanh}(uW_{\mathrm{gate}}^\top) \odot uW_{\mathrm{up}}^\top\big)W_{\mathrm{down}}^\top$ ▷ expand to 12288, gate, shrink back to 1536
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-feed-forward norm, then add residual
per-layer embedding injection
$h \leftarrow h + \mathrm{RMSNorm}\left((\mathrm{GELU}_{\tanh}(hW_{\mathrm{gate}}^{\mathrm{ple}\top}) \odot p_\ell)W_{\mathrm{proj}}^{\mathrm{ple}\top}\right)$ ▷ gate down to 256, multiply by p_ℓ, project back up
$h \leftarrow \gamma_\ell\, h$ ▷ fixed per-layer scale
return h

Algorithm 8 Block $\mathcal{B}_{\mathrm{V}}(h)$ — Vision encoder block

Require: h of shape $B \times T \times 768$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top$ ▷ $B{\times}T{\times}12{\times}64$ (12 query heads)
$K \leftarrow uW_K^\top;\ V \leftarrow uW_V^\top$ ▷ $B{\times}T{\times}12{\times}64$ (12 key/value heads)
$Q \leftarrow \mathrm{RMSNorm}(Q);\ K \leftarrow \mathrm{RMSNorm}(K)$ ▷ normalise each head vector (stabilises training)
$V \leftarrow \mathrm{RMSNorm}(V)$
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ 2-D rotary position (row, column)
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: every patch/frame sees every other
$O \leftarrow PV$ ▷ $B{\times}T{\times}12{\times}64$
$m \leftarrow O\,W_O^\top$ ▷ concatenate heads, project back to $B{\times}T{\times}768$
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-attention norm, then add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{GELU}_{\tanh}(uW_{\mathrm{gate}}^\top) \odot uW_{\mathrm{up}}^\top\big)W_{\mathrm{down}}^\top$ ▷ expand to 3072, gate, shrink back to 768
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-feed-forward norm, then add residual
return h

Algorithm 9 Block $\mathcal{B}_{\mathrm{A}}(h)$ — Conformer audio block

Require: h of shape $B \times T \times 1024$

$h \leftarrow h + \frac{1}{2}\,\mathrm{FFN}_1(h)$ ▷ first half-step feed-forward
$h \leftarrow h + \mathrm{RMSNorm}(\mathrm{Attn}_{rel}(\mathrm{RMSNorm}(h)))$ ▷ local self-attention over audio frames
$h \leftarrow h + \mathrm{ConvModule}(h)$ ▷ GLU → depthwise conv → norm → SiLU → linear
$h \leftarrow h + \frac{1}{2}\,\mathrm{FFN}_2(h)$ ▷ second half-step feed-forward
$h \leftarrow \mathrm{RMSNorm}(h)$
return h

4 Compact Architecture (repeated layers written once)

4.1 Modules outside the repeated stacks

Table 3: Weights that appear once

Weight	Type	Shape	Params
`model.vision_tower.patch_embedder.position_embedding_table`	Gemma4VisionPatchEmbedder	$2 \times 10240 \times 768$	15,728,640
`model.vision_tower.patch_embedder.input_proj.weight`	Linear	768×768	589,824
`model.language_model.embed_tokens.weight`	Gemma4TextScaledWordEmbedding	262144×1536	402,653,184
`model.language_model.norm.weight`	Gemma4RMSNorm	1536	1,536
`model.language_model.embed_tokens_per_layer.weight`	Gemma4TextScaledWordEmbedding	262144×8960	2,348,810,240
`model.language_model.per_layer_model_projection.weight`	Linear	8960×1536	13,762,560
`model.language_model.per_layer_projection_norm.weight`	Gemma4RMSNorm	256	256
`model.audio_tower.subsample_conv_projection.layer0.conv.weight`	Conv2d	$128 \times 1 \times 3 \times 3$	1,152
`model.audio_tower.subsample_conv_projection.layer0.norm.weight`	LayerNorm	128	128

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Weight	Type	Shape	Params
`model.audio_tower.subsample_conv_projection.layer1.conv.weight`	Conv2d	$32 \times 128 \times 3 \times 3$	36,864
`model.audio_tower.subsample_conv_projection.layer1.norm.weight`	LayerNorm	32	32
`model.audio_tower.subsample_conv_projection.input_proj_linear.weight`	Linear	1024×1024	1,048,576
`model.audio_tower.output_proj.weight`	Linear	1536×1024	1,572,864
`model.audio_tower.output_proj.bias`	Linear	1536	1,536
`model.embed_vision.embedding_projection.weight`	Linear	1536×768	1,179,648
`model.embed_audio.embedding_projection.weight`	Linear	1536×1536	2,359,296
`lm_head.weight` (tied to `model.language_model.embed_tokens.weight`)	Linear	262144×1536	0 (shared)

4.2 Language-model decoder: `model.language_model.layers`

Table 4: $\mathcal{B}_{\mathrm{S1}}$: Sliding-window attention block (MLP width 6144) — layers 0–3, 5–8, 10–13; weight names are relative to `model.language_model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	2048×1536	3,145,728
`self_attn.q_norm.weight`	Gemma4RMSNorm	256	256
`self_attn.k_norm.weight`	Gemma4RMSNorm	256	256
`self_attn.k_proj.weight`	Linear	256×1536	393,216
`self_attn.v_proj.weight`	Linear	256×1536	393,216
`self_attn.o_proj.weight`	Linear	1536×2048	3,145,728
`mlp.gate_proj.weight`	Linear	6144×1536	9,437,184
`mlp.up_proj.weight`	Linear	6144×1536	9,437,184
`mlp.down_proj.weight`	Linear	1536×6144	9,437,184
`input_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`post_attention_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`pre_feedforward_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`post_feedforward_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`per_layer_input_gate.weight`	Linear	256×1536	393,216
`per_layer_projection.weight`	Linear	1536×256	393,216
`post_per_layer_input_norm.weight`	Gemma4RMSNorm	1536	1,536
Per block			**36,184,064**
× 12 blocks			**434,208,768**

Table 5: $\mathcal{B}_{\mathrm{F1}}$: Full (global) attention block (MLP width 6144) — layers 4, 9, …, 14 (every 5th); weight names are relative to `model.language_model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	4096×1536	6,291,456
`self_attn.q_norm.weight`	Gemma4RMSNorm	512	512
`self_attn.k_norm.weight`	Gemma4RMSNorm	512	512
`self_attn.k_proj.weight`	Linear	512×1536	786,432
`self_attn.v_proj.weight`	Linear	512×1536	786,432
`self_attn.o_proj.weight`	Linear	1536×4096	6,291,456
`mlp.gate_proj.weight`	Linear	6144×1536	9,437,184

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Weight (relative)	Type	Shape	Params
`mlp.up_proj.weight`	Linear	6144×1536	9,437,184
`mlp.down_proj.weight`	Linear	1536×6144	9,437,184
`input_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`post_attention_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`pre_feedforward_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`post_feedforward_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`per_layer_input_gate.weight`	Linear	256×1536	393,216
`per_layer_projection.weight`	Linear	1536×256	393,216
`post_per_layer_input_norm.weight`	Gemma4RMSNorm	1536	1,536
Per block			**43,262,464**
× 3 blocks			**129,787,392**

Table 6: $\mathcal{B}_{S2}$: Sliding-window attention block (KV-shared, MLP width 12288) — layers 15–18, 20–23, 25–28, 30–33; weight names are relative to `model.language_model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	2048×1536	3,145,728
`self_attn.q_norm.weight`	Gemma4RMSNorm	256	256
`self_attn.o_proj.weight`	Linear	1536×2048	3,145,728
`mlp.gate_proj.weight`	Linear	12288×1536	18,874,368
`mlp.up_proj.weight`	Linear	12288×1536	18,874,368
`mlp.down_proj.weight`	Linear	1536×12288	18,874,368
`input_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`post_attention_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`pre_feedforward_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`post_feedforward_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`per_layer_input_gate.weight`	Linear	256×1536	393,216
`per_layer_projection.weight`	Linear	1536×256	393,216
`post_per_layer_input_norm.weight`	Gemma4RMSNorm	1536	1,536
Per block			**63,708,928**
× 16 blocks			**1,019,342,848**

Table 7: $\mathcal{B}_{F2}$: Full (global) attention block (KV-shared, MLP width 12288) — layers 19, 24, …, 34 (every 5th); weight names are relative to `model.language_model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	4096×1536	6,291,456
`self_attn.q_norm.weight`	Gemma4RMSNorm	512	512
`self_attn.o_proj.weight`	Linear	1536×4096	6,291,456
`mlp.gate_proj.weight`	Linear	12288×1536	18,874,368
`mlp.up_proj.weight`	Linear	12288×1536	18,874,368
`mlp.down_proj.weight`	Linear	1536×12288	18,874,368
`input_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`post_attention_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`pre_feedforward_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`post_feedforward_layernorm.weight`	Gemma4RMSNorm	1536	1,536
`per_layer_input_gate.weight`	Linear	256×1536	393,216
`per_layer_projection.weight`	Linear	1536×256	393,216
`post_per_layer_input_norm.weight`	Gemma4RMSNorm	1536	1,536
Per block			**70,000,640**
× 4 blocks			**280,002,560**

4.3 Vision encoder: `model.vision_tower.encoder.layers`

Table 8: $\mathcal{B}_{\mathrm{V}}$: Vision encoder block — layers 0–15; weight names are relative to `model.vision_tower.encoder.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.linear.weight`	Linear	768×768	589,824
`self_attn.k_proj.linear.weight`	Linear	768×768	589,824
`self_attn.v_proj.linear.weight`	Linear	768×768	589,824
`self_attn.o_proj.linear.weight`	Linear	768×768	589,824
`self_attn.q_norm.weight`	Gemma4RMSNorm	64	64
`self_attn.k_norm.weight`	Gemma4RMSNorm	64	64
`mlp.gate_proj.linear.weight`	Linear	3072×768	2,359,296
`mlp.up_proj.linear.weight`	Linear	3072×768	2,359,296
`mlp.down_proj.linear.weight`	Linear	768×3072	2,359,296
`input_layernorm.weight`	Gemma4RMSNorm	768	768
`post_attention_layernorm.weight`	Gemma4RMSNorm	768	768
`pre_feedforward_layernorm.weight`	Gemma4RMSNorm	768	768
`post_feedforward_layernorm.weight`	Gemma4RMSNorm	768	768
Per block			**9,440,384**
$\times$ 16 blocks			**151,046,144**

4.4 Audio encoder: `model.audio_tower.layers`

Table 9: $\mathcal{B}_{\mathrm{A}}$: Conformer audio block — layers 0–11; weight names are relative to `model.audio_tower.layers.`ℓ

Weight (relative)	Type	Shape	Params
`feed_forward1.ffw_layer_1.linear.weight`	Linear	4096×1024	4,194,304
`feed_forward1.ffw_layer_2.linear.weight`	Linear	1024×4096	4,194,304
`feed_forward1.pre_layer_norm.weight`	Gemma4RMSNorm	1024	1,024
`feed_forward1.post_layer_norm.weight`	Gemma4RMSNorm	1024	1,024
`feed_forward2.ffw_layer_1.linear.weight`	Linear	4096×1024	4,194,304
`feed_forward2.ffw_layer_2.linear.weight`	Linear	1024×4096	4,194,304
`feed_forward2.pre_layer_norm.weight`	Gemma4RMSNorm	1024	1,024
`feed_forward2.post_layer_norm.weight`	Gemma4RMSNorm	1024	1,024
`self_attn.per_dim_scale`	Gemma4AudioAttention	128	128
`self_attn.q_proj.linear.weight`	Linear	1024×1024	1,048,576
`self_attn.k_proj.linear.weight`	Linear	1024×1024	1,048,576
`self_attn.v_proj.linear.weight`	Linear	1024×1024	1,048,576
`self_attn.post.linear.weight`	Linear	1024×1024	1,048,576
`self_attn.relative_k_proj.weight`	Linear	1024×1024	1,048,576
`lconv1d.linear_start.linear.weight`	Linear	2048×1024	2,097,152
`lconv1d.linear_end.linear.weight`	Linear	1024×1024	1,048,576
`lconv1d.depthwise_conv1d.weight`	Gemma4AudioCausalConv1d	$1024 \times 1 \times 5$	5,120
`lconv1d.pre_layer_norm.weight`	Gemma4RMSNorm	1024	1,024
`lconv1d.conv_norm.weight`	Gemma4RMSNorm	1024	1,024
`norm_pre_attn.weight`	Gemma4RMSNorm	1024	1,024
`norm_post_attn.weight`	Gemma4RMSNorm	1024	1,024
`norm_out.weight`	Gemma4RMSNorm	1024	1,024
Per block			**25,180,288**

continued on next page

Weight (relative)	Type	Shape	Params
× **12 blocks**			**302,163,456**

5 What Each Layer Does

One row per *kind* of layer present in this model. "Appears as" gives the module names (layer indices removed).

Layer (appears as)	What it does
Language model	
Token embedding `embed_tokens`	Lookup table with one learned vector per vocabulary token. Turns token ids into the vectors the network computes with (Gemma multiplies them by $\sqrt{d}$).
Per-layer token embedding (PLE) `embed_tokens_per_layer`	A second embedding table looked up from the same token ids. It holds one small vector per token for every decoder layer, which gives each layer extra token-specific information cheaply. It can sit in CPU/flash memory, which is what the "E" (effective size) in the model name refers to.
PLE gate (down-projection) `layers.per_layer_input_gate`	Inside each layer: projects the hidden state down to the PLE size; after GELU it multiplies this layer's per-layer embedding p_ℓ.
PLE up-projection `layers.per_layer_projection`	Projects the gated per-layer signal back up to the hidden size before it is added to the residual stream.
PLE output norm `layers.post_per_layer_input_norm`	RMSNorm on the per-layer contribution before the residual add.
PLE context projection `per_layer_model_projection`	Projects the main token embedding into one small vector per layer. Added to the PLE lookup, so the per-layer input depends on context as well as token identity.
PLE projection norm `per_layer_projection_norm`	RMSNorm on the projected per-layer inputs.
Query projection (q_proj) `self_attn.q_proj`	Turns each token vector into query vectors, one per head: "what am I looking for?".
Query norm (q_norm) `self_attn.q_norm`	RMSNorm applied to every query head vector before RoPE. Keeps attention logits in a sane range (QK-norm).
Key norm (k_norm) `self_attn.k_norm`	RMSNorm applied to every key head vector before RoPE.
Key projection (k_proj) `self_attn.k_proj`	Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads.
Value projection (v_proj) `self_attn.v_proj`	Turns each token into value vectors: the information actually copied to tokens that attend to it.
Output projection (o_proj) `self_attn.o_proj`	Concatenates all heads' results and mixes them back into one hidden-size vector per token.
Gate projection (gate_proj) `mlp.gate_proj`	Expands the token vector to the MLP width and passes it through SiLU/GELU. The result acts as a soft on/off switch for each hidden unit.
Up projection (up_proj) `mlp.up_proj`	Expands the token vector to the MLP width (the "content" path). It is multiplied element-wise by the gate (SwiGLU/GeGLU).
Down projection (down_proj) `mlp.down_proj`	Shrinks the gated hidden vector back to the model width so it can be added to the residual stream.
Input norm (input_layernorm) `layers.input_layernorm`	RMSNorm applied to the residual stream before attention. Rescales each token vector to unit RMS, then multiplies by a learned per-channel weight.
post_attention_layernorm `layers.post_attention_layernorm`	RMSNorm applied to the attention output before it is added to the residual stream (Gemma "sandwich" norm).
Pre-MLP norm (pre_feedforward_layernorm) `layers.pre_feedforward_layernorm`	RMSNorm on the residual stream before the MLP (Gemma).

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Layer (appears as)	What it does
Post-MLP norm (post_feedforward_layernorm) `layers.post_feedforward_layernorm`	RMSNorm on the MLP output before it is added back (Gemma "sandwich" norm).
Final norm `norm`	RMSNorm applied once after the last decoder layer, just before the LM head.
VISION ENCODER	
2-D position table `vision_tower.patch_embedder`	Learned vectors for a patch's x and y coordinate, added so the encoder knows where each patch sits in the image.
Patch embedding `patch_embedder.input_proj`	Each image patch (16×16 pixels × 3 colours, flattened) is mapped linearly to one vector.
Query projection (q_proj) `self_attn.q_proj`	Turns each token vector into query vectors, one per head: "what am I looking for?".
Key projection (k_proj) `self_attn.k_proj`	Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads.
Value projection (v_proj) `self_attn.v_proj`	Turns each token into value vectors: the information actually copied to tokens that attend to it.
Output projection (o_proj) `self_attn.o_proj`	Concatenates all heads' results and mixes them back into one hidden-size vector per token.
Query norm (q_norm) `self_attn.q_norm`	RMSNorm applied to every query head vector before RoPE. Keeps attention logits in a sane range (QK-norm).
Key norm (k_norm) `self_attn.k_norm`	RMSNorm applied to every key head vector before RoPE.
Gate projection (gate_proj) `mlp.gate_proj`	Expands the token vector to the MLP width and passes it through SiLU/GELU. The result acts as a soft on/off switch for each hidden unit.
Up projection (up_proj) `mlp.up_proj`	Expands the token vector to the MLP width (the "content" path). It is multiplied element-wise by the gate (SwiGLU/GeGLU).
Down projection (down_proj) `mlp.down_proj`	Shrinks the gated hidden vector back to the model width so it can be added to the residual stream.
Input norm (input_layernorm) `layers.input_layernorm`	RMSNorm applied to the residual stream before attention. Rescales each token vector to unit RMS, then multiplies by a learned per-channel weight.
post_attention_layernorm `layers.post_attention_layernorm`	RMSNorm applied to the attention output before it is added to the residual stream (Gemma "sandwich" norm).
Pre-MLP norm (pre_feedforward_layernorm) `layers.pre_feedforward_layernorm`	RMSNorm on the residual stream before the MLP (Gemma).
Post-MLP norm (post_feedforward_layernorm) `layers.post_feedforward_layernorm`	RMSNorm on the MLP output before it is added back (Gemma "sandwich" norm).
AUDIO ENCODER	
Subsampling convolution `layer0.conv`, `layer1.conv`	2-D convolution (stride 2) over the time × mel-frequency spectrogram. Two of them shorten the time axis 4×, so the encoder works on fewer, richer frames.
Subsampling norm `layer0.norm`, `layer1.norm`	LayerNorm over channels after each subsampling convolution.
Audio input projection `subsample_conv_projection.input_proj_linear`	Flattens each frame's conv features and projects them to the audio-encoder width.
Conformer FFN up-projection `feed_forward1.ffw_layer_1`, `feed_forward2.ffw_layer_1`	Expands each frame vector (e.g. 1024→4096) before the SiLU non-linearity.
Conformer FFN down-projection `feed_forward1.ffw_layer_2`, `feed_forward2.ffw_layer_2`	Projects back to encoder width. The result is halved ("macaron" half-step) and added to the residual.

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Layer (appears as)	What it does
Conformer FFN pre-norm `feed_forward1.pre_layer_norm`, `feed_forward2.pre_layer_norm`	RMSNorm before each of the two conformer feed-forward modules.
Conformer FFN post-norm `feed_forward1.post_layer_norm`, `feed_forward2.post_layer_norm`	RMSNorm on the feed-forward output before the residual add.
Per-dimension query scale `layers.self_attn`	A learned scale per head dimension of the query, replacing the fixed $1/\sqrt{d_h}$.
Audio Q/K/V projections `self_attn.q_proj`, `self_attn.k_proj`, `self_attn.v_proj`	Turn each audio frame into a query (what am I looking for), key (what do I contain) and value (what do I pass on). "Clippable" linears clamp activations for numerical safety.
Audio attention output projection `self_attn.post`	Mixes the concatenated head outputs back to encoder width.
Relative-position key projection `self_attn.relative_k_proj`	Projects relative-distance encodings so attention scores depend on how far apart two frames are (Transformer-XL style) instead of RoPE.
Conv-module input projection (GLU) `lconv1d.linear_start`	Doubles the width; a Gated Linear Unit then uses one half of the channels to gate the other half.
Conv-module output projection `lconv1d.linear_end`	Projects the convolved features back before the residual add.
Depthwise causal 1-D convolution `lconv1d.depthwise_conv1d`	Each channel is convolved over time with its own small kernel (5 frames), picking up local acoustic patterns such as phoneme transitions.
Conv-module pre-norm `lconv1d.pre_layer_norm`	RMSNorm before the light-weight convolution module.
Conv-module norm `lconv1d.conv_norm`	RMSNorm after the depthwise convolution.
Pre-attention norm `layers.norm_pre_attn`	RMSNorm applied to frames before audio self-attention.
Post-attention norm `layers.norm_post_attn`	RMSNorm on the attention output before the residual add.
Block output norm `layers.norm_out`	RMSNorm at the end of every conformer block.
Audio output projection `audio_tower.output_proj`	Maps the last audio-encoder states to the size the audio embedder expects.
Multimodal projector	
Image-to-text projector `embed_vision.embedding_projection`	Normalises pooled vision features and projects them into the language model's hidden size ("soft image tokens").
Audio-to-text projector `embed_audio.embedding_projection`	Normalises audio-encoder outputs and projects them into the language model's hidden size, so they can take the place of text-token embeddings.

6 Parameter Count

Table 11: Parameters by component

Component	Parameters	% of total
Per-layer embeddings (PLE)	2,390,151,936	46.8%
MLP / MoE	1,557,135,360	30.5%
Token embedding	402,653,184	7.9%
Audio encoder	304,824,608	6.0%
Attention / token mixer	278,412,288	5.5%
Vision encoder	167,364,608	3.3%

continued on next page

Component	Parameters	% of total
Multimodal projector	3,538,944	0.1%
Norms	216,576	0.0%
Total	**5,104,297,504**	**100.0%**

Table 12: Parameters by block type

Block type	Count	Params / block	Total
$\mathcal{B}_{S1}$ Sliding-window attention block (MLP width 6144)	12	36,184,064	434,208,768
$\mathcal{B}_{F1}$ Full (global) attention block (MLP width 6144)	3	43,262,464	129,787,392
$\mathcal{B}_{S2}$ Sliding-window attention block (KV-shared, MLP width 12288)	16	63,708,928	1,019,342,848
$\mathcal{B}_{F2}$ Full (global) attention block (KV-shared, MLP width 12288)	4	70,000,640	280,002,560
$\mathcal{B}_{V}$ Vision encoder block	16	9,440,384	151,046,144
$\mathcal{B}_{A}$ Conformer audio block	12	25,180,288	302,163,456
Modules outside stacks	1	–	2,787,746,336
Total			**5,104,297,504**

Appendix A Full Layer-by-Layer Table

Every weight tensor with every repetition written out (the original table).

Layer path	Type	Shape	Dtype	Params
`lm_head.weight` *(tied)*	Linear	262144 × 1536	bfloat16	402,653,184
`model`	Gemma4Model	–	–	–
`model.audio_tower`	Gemma4AudioModel	–	–	–
`model.audio_tower.layers.0`	Gemma4AudioLayer	–	–	–
`model.audio_tower.layers.0.feed_forward1`	Gemma4AudioFeed Forward	–	–	–
`model.audio_tower.layers.0.feed_forward1. ffw_layer_1`	Gemma4ClippableLinear	–	–	–
`model.audio_tower.layers.0.feed_forward1. ffw_layer_1.linear.weight`	Linear	4096 × 1024	bfloat16	4,194,304
`model.audio_tower.layers.0.feed_forward1. ffw_layer_2`	Gemma4ClippableLinear	–	–	–
`model.audio_tower.layers.0.feed_forward1. ffw_layer_2.linear.weight`	Linear	1024 × 4096	bfloat16	4,194,304
`model.audio_tower.layers.0.feed_forward1. post_layer_norm.weight`	Gemma4RMSNorm	1024	bfloat16	1,024
`model.audio_tower.layers.0.feed_forward1. pre_layer_norm.weight`	Gemma4RMSNorm	1024	bfloat16	1,024
`model.audio_tower.layers.0.feed_forward2`	Gemma4AudioFeed Forward	–	–	–
`model.audio_tower.layers.0.feed_forward2. ffw_layer_1`	Gemma4ClippableLinear	–	–	–
`model.audio_tower.layers.0.feed_forward2. ffw_layer_1.linear.weight`	Linear	4096 × 1024	bfloat16	4,194,304
`model.audio_tower.layers.0.feed_forward2. ffw_layer_2`	Gemma4ClippableLinear	–	–	–
`model.audio_tower.layers.0.feed_forward2. ffw_layer_2.linear.weight`	Linear	1024 × 4096	bfloat16	4,194,304
`model.audio_tower.layers.0.feed_forward2. post_layer_norm.weight`	Gemma4RMSNorm	1024	bfloat16	1,024
`model.audio_tower.layers.0.feed_forward2. pre_layer_norm.weight`	Gemma4RMSNorm	1024	bfloat16	1,024
`model.audio_tower.layers.0.lconv1d`	Gemma4AudioLight Conv1d	–	–	–
`model.audio_tower.layers.0.lconv1d.conv_norm. weight`	Gemma4RMSNorm	1024	bfloat16	1,024
`model.audio_tower.layers.0.lconv1d. depthwise_conv1d.weight`	Gemma4AudioCausal Conv1d	1024 × 1 × 5	bfloat16	5,120
`model.audio_tower.layers.0.lconv1d.linear_end`	Gemma4ClippableLinear	–	–	–
`model.audio_tower.layers.0.lconv1d. linear_end.linear.weight`	Linear	1024 × 1024	bfloat16	1,048,576

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.0.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.0.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.0.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.0.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.0.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.0.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.0.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.0.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.0.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.0.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.0.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.0.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.0.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.0.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.0.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.0.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.0.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.1	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.1.feed_forward1	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.1.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.1.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.1.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.1.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.1.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.1.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.1.feed_forward2	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.1.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.1.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.1.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.1.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.1.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.1.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.1.lconv1d	Gemma4AudioLight Conv1d	–	–	–
model.audio_tower.layers.1.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.1.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausal Conv1d	1024 × 1 × 5	bfloat16	5,120
model.audio_tower.layers.1.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.1.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.1.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.1.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.1.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.1.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.1.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.1.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.1.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.1.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.1.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.1.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.1.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.1.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.1.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.1.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.1.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.1.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.1.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.2	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.2.feed_forward1	Gemma4AudioFeedForward	–	–	–
model.audio_tower.layers.2.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.2.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.2.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.2.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.2.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.2.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.2.feed_forward2	Gemma4AudioFeedForward	–	–	–
model.audio_tower.layers.2.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.2.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.2.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.2.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.2.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.2.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.2.lconv1d	Gemma4AudioLightConv1d	–	–	–
model.audio_tower.layers.2.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.2.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d	1024 × 1 × 5	bfloat16	5,120
model.audio_tower.layers.2.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.2.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.2.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.2.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.2.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.2.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.2.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.2.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.2.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.2.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.2.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.2.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.2.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.2.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.2.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.2.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.2.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.2.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.2.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.3	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.3.feed_forward1	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.3.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.3.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.3.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.3.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.3.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.3.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.3.feed_forward2	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.3.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.3.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.3.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.3.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.3.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.3.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.3.lconv1d	Gemma4AudioLight Conv1d	–	–	–
model.audio_tower.layers.3.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.3.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausal Conv1d	1024 × 1 × 5	bfloat16	5,120
model.audio_tower.layers.3.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.3.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.3.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.3.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.3.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.3.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.3.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.3.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.3.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.3.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.3.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.3.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.3.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.3.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.3.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.3.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.3.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.3.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.3.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.4	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.4.feed_forward1	Gemma4AudioFeedForward	–	–	–
model.audio_tower.layers.4.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.4.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.4.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.4.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.4.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.4.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.4.feed_forward2	Gemma4AudioFeedForward	–	–	–
model.audio_tower.layers.4.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.4.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.4.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.4.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.4.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.4.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.4.lconv1d	Gemma4AudioLightConv1d	–	–	–
model.audio_tower.layers.4.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.4.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d	1024 × 1 × 5	bfloat16	5,120
model.audio_tower.layers.4.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.4.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.4.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.4.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.4.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.4.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.4.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.4.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.4.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.4.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.4.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.4.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.4.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.4.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.4.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.4.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.4.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.4.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.4.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.5	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.5.feed_forward1	Gemma4AudioFeedForward	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.5.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.5.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.5.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.5.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.5.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.5.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.5.feed_forward2	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.5.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.5.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.5.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.5.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.5.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.5.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.5.lconv1d	Gemma4AudioLight Conv1d	–	–	–
model.audio_tower.layers.5.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.5.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausal Conv1d	1024 × 1 × 5	bfloat16	5,120
model.audio_tower.layers.5.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.5.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.5.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.5.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.5.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.5.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.5.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.5.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.5.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.5.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.5.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.5.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.5.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.5.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.5.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.5.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.5.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.5.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.5.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.6	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.6.feed_forward1	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.6.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.6.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.6.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.6.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.6.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.6.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.6.feed_forward2	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.6.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.6.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.6.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.6.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.6.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.6.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.6.lconv1d	Gemma4AudioLight Conv1d	–	–	–
model.audio_tower.layers.6.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.6.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausal Conv1d	1024 × 1 × 5	bfloat16	5,120
model.audio_tower.layers.6.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.6.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.6.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.6.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.6.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.6.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.6.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.6.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.6.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.6.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.6.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.6.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.6.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.6.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.6.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.6.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.6.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.6.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.6.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.7	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.7.feed_forward1	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.7.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.7.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.7.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.7.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.7.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.7.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.7.feed_forward2	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.7.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.7.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.7.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.7.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.7.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.7.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.7.lconv1d	Gemma4AudioLightConv1d	–	–	–
model.audio_tower.layers.7.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.7.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d	1024 × 1 × 5	bfloat16	5,120
model.audio_tower.layers.7.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.7.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.7.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.7.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.7.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.7.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.7.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.7.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.7.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.7.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.7.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.7.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.7.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.7.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.7.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.7.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.7.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.7.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.7.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.8	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.8.feed_forward1	Gemma4AudioFeedForward	–	–	–
model.audio_tower.layers.8.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.8.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.8.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.8.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.8.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.8.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.8.feed_forward2	Gemma4AudioFeedForward	–	–	–
model.audio_tower.layers.8.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.8.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.8.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.8.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.8.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.8.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.8.lconv1d	Gemma4AudioLightConv1d	–	–	–
model.audio_tower.layers.8.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.8.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d	1024 × 1 × 5	bfloat16	5,120
model.audio_tower.layers.8.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.8.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.8.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.8.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.8.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.8.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.8.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.8.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.8.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.8.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.8.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.8.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.8.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.8.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.8.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.8.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.8.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.8.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.8.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.9	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.9.feed_forward1	Gemma4AudioFeedForward	–	–	–
model.audio_tower.layers.9.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.9.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.9.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.9.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.9.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.9.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.9.feed_forward2	Gemma4AudioFeedForward	–	–	–
model.audio_tower.layers.9.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.9.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.9.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.9.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.9.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.9.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.9.lconv1d	Gemma4AudioLightConv1d	–	–	–
model.audio_tower.layers.9.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.9.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d	1024 × 1 × 5	bfloat16	5,120

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.9.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.9.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.9.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.9.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.9.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.9.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.9.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.9.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.9.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.9.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.9.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.9.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.9.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.9.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.9.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.9.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.9.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.9.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.9.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.10	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.10.feed_forward1	Gemma4AudioFeedForward	–	–	–
model.audio_tower.layers.10.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.10.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.10.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.10.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.10.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.10.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.10.feed_forward2	Gemma4AudioFeedForward	–	–	–
model.audio_tower.layers.10.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.10.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.10.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.10.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.10.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.10.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.10.lconv1d	Gemma4AudioLightConv1d	–	–	–
model.audio_tower.layers.10.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.10.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausalConv1d	1024 × 1 × 5	bfloat16	5,120
model.audio_tower.layers.10.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.10.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.10.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.10.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.10.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.10.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.10.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.10.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.10.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.10.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.10.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.10.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.10.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.10.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.10.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.10.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.10.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.10.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.10.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.11	Gemma4AudioLayer	–	–	–
model.audio_tower.layers.11.feed_forward1	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.11.feed_forward1.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.11.feed_forward1.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.11.feed_forward1.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.11.feed_forward1.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.11.feed_forward1.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.11.feed_forward1.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.11.feed_forward2	Gemma4AudioFeed Forward	–	–	–
model.audio_tower.layers.11.feed_forward2.ffw_layer_1	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.11.feed_forward2.ffw_layer_1.linear.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.audio_tower.layers.11.feed_forward2.ffw_layer_2	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.11.feed_forward2.ffw_layer_2.linear.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.audio_tower.layers.11.feed_forward2.post_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.11.feed_forward2.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.11.lconv1d	Gemma4AudioLight Conv1d	–	–	–
model.audio_tower.layers.11.lconv1d.conv_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.11.lconv1d.depthwise_conv1d.weight	Gemma4AudioCausal Conv1d	1024 × 1 × 5	bfloat16	5,120
model.audio_tower.layers.11.lconv1d.linear_end	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.11.lconv1d.linear_end.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.11.lconv1d.linear_start	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.11.lconv1d.linear_start.linear.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.audio_tower.layers.11.lconv1d.pre_layer_norm.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.11.norm_out.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.11.norm_post_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024
model.audio_tower.layers.11.norm_pre_attn.weight	Gemma4RMSNorm	1024	bfloat16	1,024

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Layer path	Type	Shape	Dtype	Params
model.audio_tower.layers.11.self_attn	Gemma4AudioAttention	–	–	–
model.audio_tower.layers.11.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.11.self_attn.k_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.11.self_attn.per_dim_scale	Gemma4AudioAttention	128	bfloat16	128
model.audio_tower.layers.11.self_attn.post	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.11.self_attn.post.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.11.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.11.self_attn.q_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.11.self_attn.relative_k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.layers.11.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.audio_tower.layers.11.self_attn.v_proj.linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.output_proj.bias	Linear	1536	bfloat16	1,536
model.audio_tower.output_proj.weight	Linear	1536 × 1024	bfloat16	1,572,864
model.audio_tower.subsample_conv_projection	Gemma4AudioSubSampleConvProjection	–	–	–
model.audio_tower.subsample_conv_projection.input_proj_linear.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.audio_tower.subsample_conv_projection.layer0	Gemma4AudioSubSampleConvProjectionLayer	–	–	–
model.audio_tower.subsample_conv_projection.layer0.conv.weight	Conv2d	128 × 1 × 3 × 3	bfloat16	1,152
model.audio_tower.subsample_conv_projection.layer0.norm.weight	LayerNorm	128	bfloat16	128
model.audio_tower.subsample_conv_projection.layer1	Gemma4AudioSubSampleConvProjectionLayer	–	–	–
model.audio_tower.subsample_conv_projection.layer1.conv.weight	Conv2d	32 × 128 × 3 × 3	bfloat16	36,864
model.audio_tower.subsample_conv_projection.layer1.norm.weight	LayerNorm	32	bfloat16	32
model.embed_audio	Gemma4MultimodalEmbedder	–	–	–
model.embed_audio.embedding_projection.weight	Linear	1536 × 1536	bfloat16	2,359,296
model.embed_vision	Gemma4MultimodalEmbedder	–	–	–
model.embed_vision.embedding_projection.weight	Linear	1536 × 768	bfloat16	1,179,648
model.language_model	Gemma4TextModel	–	–	–
model.language_model.embed_tokens.weight	Gemma4TextScaledWordEmbedding	262144 × 1536	bfloat16	402,653,184
model.language_model.embed_tokens_per_layer.weight	Gemma4TextScaledWordEmbedding	262144 × 8960	bfloat16	2,348,810,240
model.language_model.layers.0	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.0.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.0.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.0.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.0.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.0.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.0.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.0.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.0.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.0.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.0.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.0.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.0.self_attn	Gemma4TextAttention	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.0.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.0.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.0.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.0.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.0.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.0.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.1	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.1.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.1.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.1.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.1.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.1.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.1.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.1.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.1.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.1.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.1.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.1.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.1.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.1.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.1.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.1.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.1.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.1.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.1.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.2	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.2.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.2.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.2.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.2.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.2.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.2.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.2.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.2.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.2.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.2.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.2.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.2.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.2.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.2.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.2.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.2.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.2.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.2.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.3	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.3.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.3.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.3.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.3.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.3.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.3.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.3.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.3.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.3.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.3.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.3.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.3.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.3.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.3.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.3.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.3.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.3.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.3.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.4	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.4.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.4.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.4.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.4.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.4.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.4.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.4.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.4.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.4.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.4.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.4.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.4.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.4.self_attn.k_norm.weight	Gemma4RMSNorm	512	bfloat16	512
model.language_model.layers.4.self_attn.k_proj.weight	Linear	512 × 1536	bfloat16	786,432

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.4.self_attn.o_proj.weight	Linear	1536 × 4096	bfloat16	6,291,456
model.language_model.layers.4.self_attn.q_norm.weight	Gemma4RMSNorm	512	bfloat16	512
model.language_model.layers.4.self_attn.q_proj.weight	Linear	4096 × 1536	bfloat16	6,291,456
model.language_model.layers.4.self_attn.v_proj.weight	Linear	512 × 1536	bfloat16	786,432
model.language_model.layers.5	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.5.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.5.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.5.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.5.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.5.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.5.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.5.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.5.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.5.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.5.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.5.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.5.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.5.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.5.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.5.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.5.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.5.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.5.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.6	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.6.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.6.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.6.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.6.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.6.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.6.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.6.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.6.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.6.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.6.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.6.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.6.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.6.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.6.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.6.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.6.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.6.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.6.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.7	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.7.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.7.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.7.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.7.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.7.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.7.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.7.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.7.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.7.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.7.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.7.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.7.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.7.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.7.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.7.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.7.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.7.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.7.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.8	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.8.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.8.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.8.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.8.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.8.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.8.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.8.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.8.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.8.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.8.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.8.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.8.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.8.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.8.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.8.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.8.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.8.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.8.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.9	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.9.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.9.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.9.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.9.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.9.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.9.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.9.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.9.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.9.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.9.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.9.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.9.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.9.self_attn.k_norm.weight	Gemma4RMSNorm	512	bfloat16	512
model.language_model.layers.9.self_attn.k_proj.weight	Linear	512 × 1536	bfloat16	786,432
model.language_model.layers.9.self_attn.o_proj.weight	Linear	1536 × 4096	bfloat16	6,291,456
model.language_model.layers.9.self_attn.q_norm.weight	Gemma4RMSNorm	512	bfloat16	512
model.language_model.layers.9.self_attn.q_proj.weight	Linear	4096 × 1536	bfloat16	6,291,456
model.language_model.layers.9.self_attn.v_proj.weight	Linear	512 × 1536	bfloat16	786,432
model.language_model.layers.10	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.10.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.10.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.10.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.10.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.10.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.10.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.10.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.10.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.10.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.10.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.10.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.10.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.10.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.10.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.10.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.10.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.10.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.10.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.11	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.11.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.11.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.11.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.11.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.11.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.11.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.11.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.11.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.11.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.11.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.11.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.11.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.11.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.11.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.11.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.11.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.11.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.11.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.12	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.12.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.12.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.12.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.12.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.12.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.12.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.12.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.12.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.12.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.12.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.12.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.12.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.12.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.12.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.12.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.12.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.12.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.12.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.13	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.13.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.13.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.13.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.13.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.13.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.13.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.13.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.13.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.13.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.13.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.13.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.13.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.13.self_attn.k_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.13.self_attn.k_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.13.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.13.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.13.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.13.self_attn.v_proj.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.14	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.14.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.14.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.14.mlp.down_proj.weight	Linear	1536 × 6144	bfloat16	9,437,184
model.language_model.layers.14.mlp.gate_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.14.mlp.up_proj.weight	Linear	6144 × 1536	bfloat16	9,437,184
model.language_model.layers.14.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.14.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.14.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.14.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.14.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.14.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.14.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.14.self_attn.k_norm.weight	Gemma4RMSNorm	512	bfloat16	512
model.language_model.layers.14.self_attn.k_proj.weight	Linear	512 × 1536	bfloat16	786,432
model.language_model.layers.14.self_attn.o_proj.weight	Linear	1536 × 4096	bfloat16	6,291,456
model.language_model.layers.14.self_attn.q_norm.weight	Gemma4RMSNorm	512	bfloat16	512
model.language_model.layers.14.self_attn.q_proj.weight	Linear	4096 × 1536	bfloat16	6,291,456
model.language_model.layers.14.self_attn.v_proj.weight	Linear	512 × 1536	bfloat16	786,432
model.language_model.layers.15	Gemma4TextDecoderLayer	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.15.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.15.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.15.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.15.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.15.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.15.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.15.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.15.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.15.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.15.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.15.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.15.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.15.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.15.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.15.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.16	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.16.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.16.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.16.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.16.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.16.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.16.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.16.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.16.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.16.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.16.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.16.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.16.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.16.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.16.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.16.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.17	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.17.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.17.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.17.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.17.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.17.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.17.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.17.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.17.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.17.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.17.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.17.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.17.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.17.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.17.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.17.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.18	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.18.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.18.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.18.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.18.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.18.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.18.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.18.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.18.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.18.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.18.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.18.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.18.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.18.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.18.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.18.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.19	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.19.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.19.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.19.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.19.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.19.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.19.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.19.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.19.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.19.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.19.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.19.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.19.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.19.self_attn.o_proj.weight	Linear	1536 × 4096	bfloat16	6,291,456
model.language_model.layers.19.self_attn.q_norm.weight	Gemma4RMSNorm	512	bfloat16	512

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.19.self_attn.q_proj.weight	Linear	4096 × 1536	bfloat16	6,291,456
model.language_model.layers.20	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.20.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.20.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.20.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.20.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.20.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.20.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.20.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.20.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.20.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.20.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.20.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.20.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.20.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.20.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.20.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.21	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.21.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.21.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.21.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.21.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.21.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.21.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.21.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.21.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.21.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.21.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.21.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.21.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.21.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.21.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.21.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.22	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.22.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.22.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.22.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.22.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.22.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.22.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.22.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.22.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.22.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.22.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.22.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.22.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.22.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.22.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.22.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.23	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.23.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.23.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.23.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.23.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.23.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.23.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.23.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.23.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.23.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.23.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.23.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.23.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.23.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.23.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.23.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.24	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.24.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.24.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.24.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.24.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.24.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.24.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.24.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.24.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.24.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.24.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.24.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.24.self_attn	Gemma4TextAttention	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.24.self_attn.o_proj.weight	Linear	1536 × 4096	bfloat16	6,291,456
model.language_model.layers.24.self_attn.q_norm.weight	Gemma4RMSNorm	512	bfloat16	512
model.language_model.layers.24.self_attn.q_proj.weight	Linear	4096 × 1536	bfloat16	6,291,456
model.language_model.layers.25	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.25.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.25.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.25.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.25.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.25.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.25.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.25.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.25.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.25.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.25.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.25.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.25.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.25.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.25.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.25.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.26	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.26.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.26.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.26.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.26.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.26.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.26.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.26.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.26.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.26.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.26.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.26.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.26.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.26.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.26.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.26.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.27	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.27.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.27.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.27.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.27.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.27.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.27.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.27.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.27.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.27.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.27.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.27.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.27.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.27.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.27.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.27.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.28	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.28.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.28.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.28.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.28.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.28.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.28.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.28.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.28.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.28.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.28.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.28.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.28.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.28.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.28.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.28.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.29	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.29.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.29.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.29.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.29.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.29.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.29.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.29.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.29.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.29.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.29.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.29.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.29.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.29.self_attn.o_proj.weight	Linear	1536 × 4096	bfloat16	6,291,456
model.language_model.layers.29.self_attn.q_norm.weight	Gemma4RMSNorm	512	bfloat16	512
model.language_model.layers.29.self_attn.q_proj.weight	Linear	4096 × 1536	bfloat16	6,291,456
model.language_model.layers.30	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.30.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.30.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.30.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.30.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.30.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.30.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.30.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.30.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.30.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.30.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.30.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.30.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.30.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.30.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.30.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.31	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.31.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.31.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.31.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.31.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.31.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.31.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.31.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.31.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.31.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.31.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.31.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.31.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.31.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.31.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.31.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.32	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.32.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.32.mlp	Gemma4TextMLP	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.32.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.32.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.32.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.32.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.32.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.32.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.32.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.32.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.32.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.32.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.32.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.32.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.32.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.33	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.33.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.33.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.33.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.33.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.33.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.33.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.33.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.33.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.33.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.33.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.33.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.33.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.33.self_attn.o_proj.weight	Linear	1536 × 2048	bfloat16	3,145,728
model.language_model.layers.33.self_attn.q_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.language_model.layers.33.self_attn.q_proj.weight	Linear	2048 × 1536	bfloat16	3,145,728
model.language_model.layers.34	Gemma4TextDecoderLayer	–	–	–
model.language_model.layers.34.input_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.34.mlp	Gemma4TextMLP	–	–	–
model.language_model.layers.34.mlp.down_proj.weight	Linear	1536 × 12288	bfloat16	18,874,368
model.language_model.layers.34.mlp.gate_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.34.mlp.up_proj.weight	Linear	12288 × 1536	bfloat16	18,874,368
model.language_model.layers.34.per_layer_input_gate.weight	Linear	256 × 1536	bfloat16	393,216
model.language_model.layers.34.per_layer_projection.weight	Linear	1536 × 256	bfloat16	393,216
model.language_model.layers.34.post_attention_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.34.post_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.34.post_per_layer_input_norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.34.pre_feedforward_layernorm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.layers.34.self_attn	Gemma4TextAttention	–	–	–
model.language_model.layers.34.self_attn.o_proj.weight	Linear	1536 × 4096	bfloat16	6,291,456
model.language_model.layers.34.self_attn.q_norm.weight	Gemma4RMSNorm	512	bfloat16	512
model.language_model.layers.34.self_attn.q_proj.weight	Linear	4096 × 1536	bfloat16	6,291,456
model.language_model.norm.weight	Gemma4RMSNorm	1536	bfloat16	1,536
model.language_model.per_layer_model_projection.weight	Linear	8960 × 1536	bfloat16	13,762,560
model.language_model.per_layer_projection_norm.weight	Gemma4RMSNorm	256	bfloat16	256
model.vision_tower	Gemma4VisionModel	–	–	–
model.vision_tower.encoder	Gemma4VisionEncoder	–	–	–
model.vision_tower.encoder.layers.0	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.0.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.0.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.0.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.0.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.0.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.0.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.0.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.0.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.0.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.0.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.0.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.0.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.0.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.0.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.0.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.0.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.0.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.0.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.0.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.0.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.0.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.0.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.1	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.1.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.1.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.1.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.1.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.1.mlp.gate_proj	Gemma4ClippableLinear	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.1.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.1.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.1.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.1.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.1.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.1.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.1.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.1.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.1.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.1.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.1.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.1.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.1.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.1.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.1.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.1.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.1.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.2	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.2.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.2.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.2.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.2.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.2.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.2.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.2.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.2.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.2.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.2.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.2.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.2.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.2.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.2.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.2.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.2.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.2.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.2.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.2.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.2.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.2.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.2.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.3	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.3.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.3.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.3.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.3.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.3.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.3.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.3.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.3.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.3.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.3.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.3.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.3.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.3.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.3.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.3.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.3.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.3.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.3.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.3.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.3.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.3.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.3.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.4	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.4.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.4.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.4.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.4.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.4.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.4.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.4.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.4.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.4.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.4.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.4.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.4.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.4.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.4.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.4.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.4.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.4.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.4.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.4.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.4.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.4.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.4.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.5	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.5.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.5.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.5.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.5.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.5.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.5.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.5.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.5.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.5.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.5.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.5.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.5.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.5.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.5.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.5.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.5.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.5.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.5.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.5.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.5.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.5.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.5.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.6	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.6.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.6.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.6.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.6.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.6.mlp.gate_proj	Gemma4ClippableLinear	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.6.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.6.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.6.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.6.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.6.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.6.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.6.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.6.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.6.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.6.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.6.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.6.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.6.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.6.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.6.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.6.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.6.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.7	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.7.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.7.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.7.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.7.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.7.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.7.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.7.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.7.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.7.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.7.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.7.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.7.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.7.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.7.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.7.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.7.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.7.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.7.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.7.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.7.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.7.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.7.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.8	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.8.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.8.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.8.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.8.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.8.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.8.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.8.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.8.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.8.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.8.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.8.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.8.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.8.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.8.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.8.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.8.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.8.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.8.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.8.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.8.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.8.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.8.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.9	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.9.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.9.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.9.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.9.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.9.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.9.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.9.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.9.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.9.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.9.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.9.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.9.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.9.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.9.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.9.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.9.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.9.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.9.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.9.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.9.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.9.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.9.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.10	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.10.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.10.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.10.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.10.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.10.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.10.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.10.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.10.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.10.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.10.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.10.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.10.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.10.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.10.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.10.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.10.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.10.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.10.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.10.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.10.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.10.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.10.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.11	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.11.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.11.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.11.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.11.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.11.mlp.gate_proj	Gemma4ClippableLinear	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.11.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.11.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.11.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.11.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.11.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.11.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.11.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.11.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.11.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.11.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.11.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.11.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.11.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.11.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.11.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.11.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.11.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.12	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.12.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.12.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.12.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.12.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.12.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.12.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.12.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.12.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.12.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.12.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.12.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.12.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.12.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.12.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.12.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.12.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.12.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.12.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.12.self_attn.q_proj	Gemma4ClippableLinear	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.12.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.12.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.12.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.13	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.13.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.13.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.13.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.13.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.13.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.13.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.13.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.13.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.13.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.13.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.13.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.13.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.13.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.13.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.13.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.13.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.13.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.13.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.13.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.13.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.13.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.13.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.14	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.14.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.14.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.14.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.14.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.14.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.14.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.14.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.14.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.14.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.14.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.14.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.14.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.14.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.14.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.14.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.14.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.14.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.14.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.14.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.14.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.14.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.14.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.15	Gemma4VisionEncoderLayer	–	–	–
model.vision_tower.encoder.layers.15.input_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.15.mlp	Gemma4VisionMLP	–	–	–
model.vision_tower.encoder.layers.15.mlp.down_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.15.mlp.down_proj.linear.weight	Linear	768 × 3072	bfloat16	2,359,296
model.vision_tower.encoder.layers.15.mlp.gate_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.15.mlp.gate_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.15.mlp.up_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.15.mlp.up_proj.linear.weight	Linear	3072 × 768	bfloat16	2,359,296
model.vision_tower.encoder.layers.15.post_attention_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.15.post_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.15.pre_feedforward_layernorm.weight	Gemma4RMSNorm	768	bfloat16	768
model.vision_tower.encoder.layers.15.self_attn	Gemma4VisionAttention	–	–	–
model.vision_tower.encoder.layers.15.self_attn.k_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.15.self_attn.k_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.15.self_attn.k_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.15.self_attn.o_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.15.self_attn.o_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.15.self_attn.q_norm.weight	Gemma4RMSNorm	64	bfloat16	64
model.vision_tower.encoder.layers.15.self_attn.q_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.15.self_attn.q_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.encoder.layers.15.self_attn.v_proj	Gemma4ClippableLinear	–	–	–
model.vision_tower.encoder.layers.15.self_attn.v_proj.linear.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.patch_embedder	Gemma4VisionPatchEmbedder	–	–	–
model.vision_tower.patch_embedder.input_proj.weight	Linear	768 × 768	bfloat16	589,824
model.vision_tower.patch_embedder.position_embedding_table	Gemma4VisionPatchEmbedder	2 × 10240 × 768	bfloat16	15,728,640

Appendix B Per-Layer Parameter Count

Module	Params	Running total
model.audio_tower.layers.0.feed_forward1.ffw_layer_1.linear	4,194,304	4,194,304
model.audio_tower.layers.0.feed_forward1.ffw_layer_2.linear	4,194,304	8,388,608
model.audio_tower.layers.0.feed_forward1.post_layer_norm	1,024	8,389,632
model.audio_tower.layers.0.feed_forward1.pre_layer_norm	1,024	8,390,656
model.audio_tower.layers.0.feed_forward2.ffw_layer_1.linear	4,194,304	12,584,960
model.audio_tower.layers.0.feed_forward2.ffw_layer_2.linear	4,194,304	16,779,264
model.audio_tower.layers.0.feed_forward2.post_layer_norm	1,024	16,780,288
model.audio_tower.layers.0.feed_forward2.pre_layer_norm	1,024	16,781,312
model.audio_tower.layers.0.lconv1d.conv_norm	1,024	16,782,336
model.audio_tower.layers.0.lconv1d.depthwise_conv1d	5,120	16,787,456
model.audio_tower.layers.0.lconv1d.linear_end.linear	1,048,576	17,836,032
model.audio_tower.layers.0.lconv1d.linear_start.linear	2,097,152	19,933,184
model.audio_tower.layers.0.lconv1d.pre_layer_norm	1,024	19,934,208
model.audio_tower.layers.0.norm_out	1,024	19,935,232
model.audio_tower.layers.0.norm_post_attn	1,024	19,936,256
model.audio_tower.layers.0.norm_pre_attn	1,024	19,937,280
model.audio_tower.layers.0.self_attn.k_proj.linear	1,048,576	20,985,856
model.audio_tower.layers.0.self_attn	128	20,985,984
model.audio_tower.layers.0.self_attn.post.linear	1,048,576	22,034,560
model.audio_tower.layers.0.self_attn.q_proj.linear	1,048,576	23,083,136
model.audio_tower.layers.0.self_attn.relative_k_proj	1,048,576	24,131,712
model.audio_tower.layers.0.self_attn.v_proj.linear	1,048,576	25,180,288
model.audio_tower.layers.1.feed_forward1.ffw_layer_1.linear	4,194,304	29,374,592
model.audio_tower.layers.1.feed_forward1.ffw_layer_2.linear	4,194,304	33,568,896
model.audio_tower.layers.1.feed_forward1.post_layer_norm	1,024	33,569,920
model.audio_tower.layers.1.feed_forward1.pre_layer_norm	1,024	33,570,944
model.audio_tower.layers.1.feed_forward2.ffw_layer_1.linear	4,194,304	37,765,248
model.audio_tower.layers.1.feed_forward2.ffw_layer_2.linear	4,194,304	41,959,552
model.audio_tower.layers.1.feed_forward2.post_layer_norm	1,024	41,960,576
model.audio_tower.layers.1.feed_forward2.pre_layer_norm	1,024	41,961,600
model.audio_tower.layers.1.lconv1d.conv_norm	1,024	41,962,624
model.audio_tower.layers.1.lconv1d.depthwise_conv1d	5,120	41,967,744
model.audio_tower.layers.1.lconv1d.linear_end.linear	1,048,576	43,016,320
model.audio_tower.layers.1.lconv1d.linear_start.linear	2,097,152	45,113,472
model.audio_tower.layers.1.lconv1d.pre_layer_norm	1,024	45,114,496

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Module	Params	Running total
model.audio_tower.layers.1.norm_out	1,024	45,115,520
model.audio_tower.layers.1.norm_post_attn	1,024	45,116,544
model.audio_tower.layers.1.norm_pre_attn	1,024	45,117,568
model.audio_tower.layers.1.self_attn.k_proj.linear	1,048,576	46,166,144
model.audio_tower.layers.1.self_attn	128	46,166,272
model.audio_tower.layers.1.self_attn.post.linear	1,048,576	47,214,848
model.audio_tower.layers.1.self_attn.q_proj.linear	1,048,576	48,263,424
model.audio_tower.layers.1.self_attn. relative_k_proj	1,048,576	49,312,000
model.audio_tower.layers.1.self_attn.v_proj.linear	1,048,576	50,360,576
model.audio_tower.layers.2.feed_forward1. ffw_layer_1.linear	4,194,304	54,554,880
model.audio_tower.layers.2.feed_forward1. ffw_layer_2.linear	4,194,304	58,749,184
model.audio_tower.layers.2.feed_forward1. post_layer_norm	1,024	58,750,208
model.audio_tower.layers.2.feed_forward1. pre_layer_norm	1,024	58,751,232
model.audio_tower.layers.2.feed_forward2. ffw_layer_1.linear	4,194,304	62,945,536
model.audio_tower.layers.2.feed_forward2. ffw_layer_2.linear	4,194,304	67,139,840
model.audio_tower.layers.2.feed_forward2. post_layer_norm	1,024	67,140,864
model.audio_tower.layers.2.feed_forward2. pre_layer_norm	1,024	67,141,888
model.audio_tower.layers.2.lconv1d.conv_norm	1,024	67,142,912
model.audio_tower.layers.2.lconv1d.depthwise_conv1d	5,120	67,148,032
model.audio_tower.layers.2.lconv1d.linear_end. linear	1,048,576	68,196,608
model.audio_tower.layers.2.lconv1d.linear_start. linear	2,097,152	70,293,760
model.audio_tower.layers.2.lconv1d.pre_layer_norm	1,024	70,294,784
model.audio_tower.layers.2.norm_out	1,024	70,295,808
model.audio_tower.layers.2.norm_post_attn	1,024	70,296,832
model.audio_tower.layers.2.norm_pre_attn	1,024	70,297,856
model.audio_tower.layers.2.self_attn.k_proj.linear	1,048,576	71,346,432
model.audio_tower.layers.2.self_attn	128	71,346,560
model.audio_tower.layers.2.self_attn.post.linear	1,048,576	72,395,136
model.audio_tower.layers.2.self_attn.q_proj.linear	1,048,576	73,443,712
model.audio_tower.layers.2.self_attn. relative_k_proj	1,048,576	74,492,288
model.audio_tower.layers.2.self_attn.v_proj.linear	1,048,576	75,540,864
model.audio_tower.layers.3.feed_forward1. ffw_layer_1.linear	4,194,304	79,735,168
model.audio_tower.layers.3.feed_forward1. ffw_layer_2.linear	4,194,304	83,929,472
model.audio_tower.layers.3.feed_forward1. post_layer_norm	1,024	83,930,496
model.audio_tower.layers.3.feed_forward1. pre_layer_norm	1,024	83,931,520
model.audio_tower.layers.3.feed_forward2. ffw_layer_1.linear	4,194,304	88,125,824
model.audio_tower.layers.3.feed_forward2. ffw_layer_2.linear	4,194,304	92,320,128
model.audio_tower.layers.3.feed_forward2. post_layer_norm	1,024	92,321,152

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Module	Params	Running total
model.audio_tower.layers.3.feed_forward2.pre_layer_norm	1,024	92,322,176
model.audio_tower.layers.3.lconv1d.conv_norm	1,024	92,323,200
model.audio_tower.layers.3.lconv1d.depthwise_conv1d	5,120	92,328,320
model.audio_tower.layers.3.lconv1d.linear_end.linear	1,048,576	93,376,896
model.audio_tower.layers.3.lconv1d.linear_start.linear	2,097,152	95,474,048
model.audio_tower.layers.3.lconv1d.pre_layer_norm	1,024	95,475,072
model.audio_tower.layers.3.norm_out	1,024	95,476,096
model.audio_tower.layers.3.norm_post_attn	1,024	95,477,120
model.audio_tower.layers.3.norm_pre_attn	1,024	95,478,144
model.audio_tower.layers.3.self_attn.k_proj.linear	1,048,576	96,526,720
model.audio_tower.layers.3.self_attn	128	96,526,848
model.audio_tower.layers.3.self_attn.post.linear	1,048,576	97,575,424
model.audio_tower.layers.3.self_attn.q_proj.linear	1,048,576	98,624,000
model.audio_tower.layers.3.self_attn.relative_k_proj	1,048,576	99,672,576
model.audio_tower.layers.3.self_attn.v_proj.linear	1,048,576	100,721,152
model.audio_tower.layers.4.feed_forward1.ffw_layer_1.linear	4,194,304	104,915,456
model.audio_tower.layers.4.feed_forward1.ffw_layer_2.linear	4,194,304	109,109,760
model.audio_tower.layers.4.feed_forward1.post_layer_norm	1,024	109,110,784
model.audio_tower.layers.4.feed_forward1.pre_layer_norm	1,024	109,111,808
model.audio_tower.layers.4.feed_forward2.ffw_layer_1.linear	4,194,304	113,306,112
model.audio_tower.layers.4.feed_forward2.ffw_layer_2.linear	4,194,304	117,500,416
model.audio_tower.layers.4.feed_forward2.post_layer_norm	1,024	117,501,440
model.audio_tower.layers.4.feed_forward2.pre_layer_norm	1,024	117,502,464
model.audio_tower.layers.4.lconv1d.conv_norm	1,024	117,503,488
model.audio_tower.layers.4.lconv1d.depthwise_conv1d	5,120	117,508,608
model.audio_tower.layers.4.lconv1d.linear_end.linear	1,048,576	118,557,184
model.audio_tower.layers.4.lconv1d.linear_start.linear	2,097,152	120,654,336
model.audio_tower.layers.4.lconv1d.pre_layer_norm	1,024	120,655,360
model.audio_tower.layers.4.norm_out	1,024	120,656,384
model.audio_tower.layers.4.norm_post_attn	1,024	120,657,408
model.audio_tower.layers.4.norm_pre_attn	1,024	120,658,432
model.audio_tower.layers.4.self_attn.k_proj.linear	1,048,576	121,707,008
model.audio_tower.layers.4.self_attn	128	121,707,136
model.audio_tower.layers.4.self_attn.post.linear	1,048,576	122,755,712
model.audio_tower.layers.4.self_attn.q_proj.linear	1,048,576	123,804,288
model.audio_tower.layers.4.self_attn.relative_k_proj	1,048,576	124,852,864
model.audio_tower.layers.4.self_attn.v_proj.linear	1,048,576	125,901,440
model.audio_tower.layers.5.feed_forward1.ffw_layer_1.linear	4,194,304	130,095,744
model.audio_tower.layers.5.feed_forward1.ffw_layer_2.linear	4,194,304	134,290,048
model.audio_tower.layers.5.feed_forward1.post_layer_norm	1,024	134,291,072

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Module	Params	Running total
model.audio_tower.layers.5.feed_forward1.pre_layer_norm	1,024	134,292,096
model.audio_tower.layers.5.feed_forward2.ffw_layer_1.linear	4,194,304	138,486,400
model.audio_tower.layers.5.feed_forward2.ffw_layer_2.linear	4,194,304	142,680,704
model.audio_tower.layers.5.feed_forward2.post_layer_norm	1,024	142,681,728
model.audio_tower.layers.5.feed_forward2.pre_layer_norm	1,024	142,682,752
model.audio_tower.layers.5.lconv1d.conv_norm	1,024	142,683,776
model.audio_tower.layers.5.lconv1d.depthwise_conv1d	5,120	142,688,896
model.audio_tower.layers.5.lconv1d.linear_end.linear	1,048,576	143,737,472
model.audio_tower.layers.5.lconv1d.linear_start.linear	2,097,152	145,834,624
model.audio_tower.layers.5.lconv1d.pre_layer_norm	1,024	145,835,648
model.audio_tower.layers.5.norm_out	1,024	145,836,672
model.audio_tower.layers.5.norm_post_attn	1,024	145,837,696
model.audio_tower.layers.5.norm_pre_attn	1,024	145,838,720
model.audio_tower.layers.5.self_attn.k_proj.linear	1,048,576	146,887,296
model.audio_tower.layers.5.self_attn	128	146,887,424
model.audio_tower.layers.5.self_attn.post.linear	1,048,576	147,936,000
model.audio_tower.layers.5.self_attn.q_proj.linear	1,048,576	148,984,576
model.audio_tower.layers.5.self_attn.relative_k_proj	1,048,576	150,033,152
model.audio_tower.layers.5.self_attn.v_proj.linear	1,048,576	151,081,728
model.audio_tower.layers.6.feed_forward1.ffw_layer_1.linear	4,194,304	155,276,032
model.audio_tower.layers.6.feed_forward1.ffw_layer_2.linear	4,194,304	159,470,336
model.audio_tower.layers.6.feed_forward1.post_layer_norm	1,024	159,471,360
model.audio_tower.layers.6.feed_forward1.pre_layer_norm	1,024	159,472,384
model.audio_tower.layers.6.feed_forward2.ffw_layer_1.linear	4,194,304	163,666,688
model.audio_tower.layers.6.feed_forward2.ffw_layer_2.linear	4,194,304	167,860,992
model.audio_tower.layers.6.feed_forward2.post_layer_norm	1,024	167,862,016
model.audio_tower.layers.6.feed_forward2.pre_layer_norm	1,024	167,863,040
model.audio_tower.layers.6.lconv1d.conv_norm	1,024	167,864,064
model.audio_tower.layers.6.lconv1d.depthwise_conv1d	5,120	167,869,184
model.audio_tower.layers.6.lconv1d.linear_end.linear	1,048,576	168,917,760
model.audio_tower.layers.6.lconv1d.linear_start.linear	2,097,152	171,014,912
model.audio_tower.layers.6.lconv1d.pre_layer_norm	1,024	171,015,936
model.audio_tower.layers.6.norm_out	1,024	171,016,960
model.audio_tower.layers.6.norm_post_attn	1,024	171,017,984
model.audio_tower.layers.6.norm_pre_attn	1,024	171,019,008
model.audio_tower.layers.6.self_attn.k_proj.linear	1,048,576	172,067,584
model.audio_tower.layers.6.self_attn	128	172,067,712
model.audio_tower.layers.6.self_attn.post.linear	1,048,576	173,116,288
model.audio_tower.layers.6.self_attn.q_proj.linear	1,048,576	174,164,864

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Module	Params	Running total
model.audio_tower.layers.6.self_attn.relative_k_proj	1,048,576	175,213,440
model.audio_tower.layers.6.self_attn.v_proj.linear	1,048,576	176,262,016
model.audio_tower.layers.7.feed_forward1.ffw_layer_1.linear	4,194,304	180,456,320
model.audio_tower.layers.7.feed_forward1.ffw_layer_2.linear	4,194,304	184,650,624
model.audio_tower.layers.7.feed_forward1.post_layer_norm	1,024	184,651,648
model.audio_tower.layers.7.feed_forward1.pre_layer_norm	1,024	184,652,672
model.audio_tower.layers.7.feed_forward2.ffw_layer_1.linear	4,194,304	188,846,976
model.audio_tower.layers.7.feed_forward2.ffw_layer_2.linear	4,194,304	193,041,280
model.audio_tower.layers.7.feed_forward2.post_layer_norm	1,024	193,042,304
model.audio_tower.layers.7.feed_forward2.pre_layer_norm	1,024	193,043,328
model.audio_tower.layers.7.lconv1d.conv_norm	1,024	193,044,352
model.audio_tower.layers.7.lconv1d.depthwise_conv1d	5,120	193,049,472
model.audio_tower.layers.7.lconv1d.linear_end.linear	1,048,576	194,098,048
model.audio_tower.layers.7.lconv1d.linear_start.linear	2,097,152	196,195,200
model.audio_tower.layers.7.lconv1d.pre_layer_norm	1,024	196,196,224
model.audio_tower.layers.7.norm_out	1,024	196,197,248
model.audio_tower.layers.7.norm_post_attn	1,024	196,198,272
model.audio_tower.layers.7.norm_pre_attn	1,024	196,199,296
model.audio_tower.layers.7.self_attn.k_proj.linear	1,048,576	197,247,872
model.audio_tower.layers.7.self_attn	128	197,248,000
model.audio_tower.layers.7.self_attn.post.linear	1,048,576	198,296,576
model.audio_tower.layers.7.self_attn.q_proj.linear	1,048,576	199,345,152
model.audio_tower.layers.7.self_attn.relative_k_proj	1,048,576	200,393,728
model.audio_tower.layers.7.self_attn.v_proj.linear	1,048,576	201,442,304
model.audio_tower.layers.8.feed_forward1.ffw_layer_1.linear	4,194,304	205,636,608
model.audio_tower.layers.8.feed_forward1.ffw_layer_2.linear	4,194,304	209,830,912
model.audio_tower.layers.8.feed_forward1.post_layer_norm	1,024	209,831,936
model.audio_tower.layers.8.feed_forward1.pre_layer_norm	1,024	209,832,960
model.audio_tower.layers.8.feed_forward2.ffw_layer_1.linear	4,194,304	214,027,264
model.audio_tower.layers.8.feed_forward2.ffw_layer_2.linear	4,194,304	218,221,568
model.audio_tower.layers.8.feed_forward2.post_layer_norm	1,024	218,222,592
model.audio_tower.layers.8.feed_forward2.pre_layer_norm	1,024	218,223,616
model.audio_tower.layers.8.lconv1d.conv_norm	1,024	218,224,640
model.audio_tower.layers.8.lconv1d.depthwise_conv1d	5,120	218,229,760
model.audio_tower.layers.8.lconv1d.linear_end.linear	1,048,576	219,278,336
model.audio_tower.layers.8.lconv1d.linear_start.linear	2,097,152	221,375,488

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Module	Params	Running total
model.audio_tower.layers.8.lconv1d.pre_layer_norm	1,024	221,376,512
model.audio_tower.layers.8.norm_out	1,024	221,377,536
model.audio_tower.layers.8.norm_post_attn	1,024	221,378,560
model.audio_tower.layers.8.norm_pre_attn	1,024	221,379,584
model.audio_tower.layers.8.self_attn.k_proj.linear	1,048,576	222,428,160
model.audio_tower.layers.8.self_attn	128	222,428,288
model.audio_tower.layers.8.self_attn.post.linear	1,048,576	223,476,864
model.audio_tower.layers.8.self_attn.q_proj.linear	1,048,576	224,525,440
model.audio_tower.layers.8.self_attn. relative_k_proj	1,048,576	225,574,016
model.audio_tower.layers.8.self_attn.v_proj.linear	1,048,576	226,622,592
model.audio_tower.layers.9.feed_forward1. ffw_layer_1.linear	4,194,304	230,816,896
model.audio_tower.layers.9.feed_forward1. ffw_layer_2.linear	4,194,304	235,011,200
model.audio_tower.layers.9.feed_forward1. post_layer_norm	1,024	235,012,224
model.audio_tower.layers.9.feed_forward1. pre_layer_norm	1,024	235,013,248
model.audio_tower.layers.9.feed_forward2. ffw_layer_1.linear	4,194,304	239,207,552
model.audio_tower.layers.9.feed_forward2. ffw_layer_2.linear	4,194,304	243,401,856
model.audio_tower.layers.9.feed_forward2. post_layer_norm	1,024	243,402,880
model.audio_tower.layers.9.feed_forward2. pre_layer_norm	1,024	243,403,904
model.audio_tower.layers.9.lconv1d.conv_norm	1,024	243,404,928
model.audio_tower.layers.9.lconv1d.depthwise_conv1d	5,120	243,410,048
model.audio_tower.layers.9.lconv1d.linear_end. linear	1,048,576	244,458,624
model.audio_tower.layers.9.lconv1d.linear_start. linear	2,097,152	246,555,776
model.audio_tower.layers.9.lconv1d.pre_layer_norm	1,024	246,556,800
model.audio_tower.layers.9.norm_out	1,024	246,557,824
model.audio_tower.layers.9.norm_post_attn	1,024	246,558,848
model.audio_tower.layers.9.norm_pre_attn	1,024	246,559,872
model.audio_tower.layers.9.self_attn.k_proj.linear	1,048,576	247,608,448
model.audio_tower.layers.9.self_attn	128	247,608,576
model.audio_tower.layers.9.self_attn.post.linear	1,048,576	248,657,152
model.audio_tower.layers.9.self_attn.q_proj.linear	1,048,576	249,705,728
model.audio_tower.layers.9.self_attn. relative_k_proj	1,048,576	250,754,304
model.audio_tower.layers.9.self_attn.v_proj.linear	1,048,576	251,802,880
model.audio_tower.layers.10.feed_forward1. ffw_layer_1.linear	4,194,304	255,997,184
model.audio_tower.layers.10.feed_forward1. ffw_layer_2.linear	4,194,304	260,191,488
model.audio_tower.layers.10.feed_forward1. post_layer_norm	1,024	260,192,512
model.audio_tower.layers.10.feed_forward1. pre_layer_norm	1,024	260,193,536
model.audio_tower.layers.10.feed_forward2. ffw_layer_1.linear	4,194,304	264,387,840
model.audio_tower.layers.10.feed_forward2. ffw_layer_2.linear	4,194,304	268,582,144
model.audio_tower.layers.10.feed_forward2. post_layer_norm	1,024	268,583,168

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Module	Params	Running total
model.audio_tower.layers.10.feed_forward2.pre_layer_norm	1,024	268,584,192
model.audio_tower.layers.10.lconv1d.conv_norm	1,024	268,585,216
model.audio_tower.layers.10.lconv1d.depthwise_conv1d	5,120	268,590,336
model.audio_tower.layers.10.lconv1d.linear_end.linear	1,048,576	269,638,912
model.audio_tower.layers.10.lconv1d.linear_start.linear	2,097,152	271,736,064
model.audio_tower.layers.10.lconv1d.pre_layer_norm	1,024	271,737,088
model.audio_tower.layers.10.norm_out	1,024	271,738,112
model.audio_tower.layers.10.norm_post_attn	1,024	271,739,136
model.audio_tower.layers.10.norm_pre_attn	1,024	271,740,160
model.audio_tower.layers.10.self_attn.k_proj.linear	1,048,576	272,788,736
model.audio_tower.layers.10.self_attn	128	272,788,864
model.audio_tower.layers.10.self_attn.post.linear	1,048,576	273,837,440
model.audio_tower.layers.10.self_attn.q_proj.linear	1,048,576	274,886,016
model.audio_tower.layers.10.self_attn.relative_k_proj	1,048,576	275,934,592
model.audio_tower.layers.10.self_attn.v_proj.linear	1,048,576	276,983,168
model.audio_tower.layers.11.feed_forward1.ffw_layer_1.linear	4,194,304	281,177,472
model.audio_tower.layers.11.feed_forward1.ffw_layer_2.linear	4,194,304	285,371,776
model.audio_tower.layers.11.feed_forward1.post_layer_norm	1,024	285,372,800
model.audio_tower.layers.11.feed_forward1.pre_layer_norm	1,024	285,373,824
model.audio_tower.layers.11.feed_forward2.ffw_layer_1.linear	4,194,304	289,568,128
model.audio_tower.layers.11.feed_forward2.ffw_layer_2.linear	4,194,304	293,762,432
model.audio_tower.layers.11.feed_forward2.post_layer_norm	1,024	293,763,456
model.audio_tower.layers.11.feed_forward2.pre_layer_norm	1,024	293,764,480
model.audio_tower.layers.11.lconv1d.conv_norm	1,024	293,765,504
model.audio_tower.layers.11.lconv1d.depthwise_conv1d	5,120	293,770,624
model.audio_tower.layers.11.lconv1d.linear_end.linear	1,048,576	294,819,200
model.audio_tower.layers.11.lconv1d.linear_start.linear	2,097,152	296,916,352
model.audio_tower.layers.11.lconv1d.pre_layer_norm	1,024	296,917,376
model.audio_tower.layers.11.norm_out	1,024	296,918,400
model.audio_tower.layers.11.norm_post_attn	1,024	296,919,424
model.audio_tower.layers.11.norm_pre_attn	1,024	296,920,448
model.audio_tower.layers.11.self_attn.k_proj.linear	1,048,576	297,969,024
model.audio_tower.layers.11.self_attn	128	297,969,152
model.audio_tower.layers.11.self_attn.post.linear	1,048,576	299,017,728
model.audio_tower.layers.11.self_attn.q_proj.linear	1,048,576	300,066,304
model.audio_tower.layers.11.self_attn.relative_k_proj	1,048,576	301,114,880
model.audio_tower.layers.11.self_attn.v_proj.linear	1,048,576	302,163,456
model.audio_tower.output_proj	1,574,400	303,737,856
model.audio_tower.subsample_conv_projection.input_proj_linear	1,048,576	304,786,432

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Module	Params	Running total
model.audio_tower.subsample_conv_projection.layer0.conv	1,152	304,787,584
model.audio_tower.subsample_conv_projection.layer0.norm	128	304,787,712
model.audio_tower.subsample_conv_projection.layer1.conv	36,864	304,824,576
model.audio_tower.subsample_conv_projection.layer1.norm	32	304,824,608
model.embed_audio.embedding_projection	2,359,296	307,183,904
model.embed_vision.embedding_projection	1,179,648	308,363,552
model.language_model.embed_tokens	402,653,184	711,016,736
model.language_model.embed_tokens_per_layer	2,348,810,240	3,059,826,976
model.language_model.layers.0.input_layernorm	1,536	3,059,828,512
model.language_model.layers.0.mlp.down_proj	9,437,184	3,069,265,696
model.language_model.layers.0.mlp.gate_proj	9,437,184	3,078,702,880
model.language_model.layers.0.mlp.up_proj	9,437,184	3,088,140,064
model.language_model.layers.0.per_layer_input_gate	393,216	3,088,533,280
model.language_model.layers.0.per_layer_projection	393,216	3,088,926,496
model.language_model.layers.0.post_attention_layernorm	1,536	3,088,928,032
model.language_model.layers.0.post_feedforward_layernorm	1,536	3,088,929,568
model.language_model.layers.0.post_per_layer_input_norm	1,536	3,088,931,104
model.language_model.layers.0.pre_feedforward_layernorm	1,536	3,088,932,640
model.language_model.layers.0.self_attn.k_norm	256	3,088,932,896
model.language_model.layers.0.self_attn.k_proj	393,216	3,089,326,112
model.language_model.layers.0.self_attn.o_proj	3,145,728	3,092,471,840
model.language_model.layers.0.self_attn.q_norm	256	3,092,472,096
model.language_model.layers.0.self_attn.q_proj	3,145,728	3,095,617,824
model.language_model.layers.0.self_attn.v_proj	393,216	3,096,011,040
model.language_model.layers.1.input_layernorm	1,536	3,096,012,576
model.language_model.layers.1.mlp.down_proj	9,437,184	3,105,449,760
model.language_model.layers.1.mlp.gate_proj	9,437,184	3,114,886,944
model.language_model.layers.1.mlp.up_proj	9,437,184	3,124,324,128
model.language_model.layers.1.per_layer_input_gate	393,216	3,124,717,344
model.language_model.layers.1.per_layer_projection	393,216	3,125,110,560
model.language_model.layers.1.post_attention_layernorm	1,536	3,125,112,096
model.language_model.layers.1.post_feedforward_layernorm	1,536	3,125,113,632
model.language_model.layers.1.post_per_layer_input_norm	1,536	3,125,115,168
model.language_model.layers.1.pre_feedforward_layernorm	1,536	3,125,116,704
model.language_model.layers.1.self_attn.k_norm	256	3,125,116,960
model.language_model.layers.1.self_attn.k_proj	393,216	3,125,510,176
model.language_model.layers.1.self_attn.o_proj	3,145,728	3,128,655,904
model.language_model.layers.1.self_attn.q_norm	256	3,128,656,160
model.language_model.layers.1.self_attn.q_proj	3,145,728	3,131,801,888
model.language_model.layers.1.self_attn.v_proj	393,216	3,132,195,104
model.language_model.layers.2.input_layernorm	1,536	3,132,196,640
model.language_model.layers.2.mlp.down_proj	9,437,184	3,141,633,824
model.language_model.layers.2.mlp.gate_proj	9,437,184	3,151,071,008
model.language_model.layers.2.mlp.up_proj	9,437,184	3,160,508,192
model.language_model.layers.2.per_layer_input_gate	393,216	3,160,901,408
model.language_model.layers.2.per_layer_projection	393,216	3,161,294,624

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Module	Params	Running total
model.language_model.layers.2. post_attention_layernorm	1,536	3,161,296,160
model.language_model.layers.2. post_feedforward_layernorm	1,536	3,161,297,696
model.language_model.layers.2. post_per_layer_input_norm	1,536	3,161,299,232
model.language_model.layers.2. pre_feedforward_layernorm	1,536	3,161,300,768
model.language_model.layers.2.self_attn.k_norm	256	3,161,301,024
model.language_model.layers.2.self_attn.k_proj	393,216	3,161,694,240
model.language_model.layers.2.self_attn.o_proj	3,145,728	3,164,839,968
model.language_model.layers.2.self_attn.q_norm	256	3,164,840,224
model.language_model.layers.2.self_attn.q_proj	3,145,728	3,167,985,952
model.language_model.layers.2.self_attn.v_proj	393,216	3,168,379,168
model.language_model.layers.3.input_layernorm	1,536	3,168,380,704
model.language_model.layers.3.mlp.down_proj	9,437,184	3,177,817,888
model.language_model.layers.3.mlp.gate_proj	9,437,184	3,187,255,072
model.language_model.layers.3.mlp.up_proj	9,437,184	3,196,692,256
model.language_model.layers.3.per_layer_input_gate	393,216	3,197,085,472
model.language_model.layers.3.per_layer_projection	393,216	3,197,478,688
model.language_model.layers.3. post_attention_layernorm	1,536	3,197,480,224
model.language_model.layers.3. post_feedforward_layernorm	1,536	3,197,481,760
model.language_model.layers.3. post_per_layer_input_norm	1,536	3,197,483,296
model.language_model.layers.3. pre_feedforward_layernorm	1,536	3,197,484,832
model.language_model.layers.3.self_attn.k_norm	256	3,197,485,088
model.language_model.layers.3.self_attn.k_proj	393,216	3,197,878,304
model.language_model.layers.3.self_attn.o_proj	3,145,728	3,201,024,032
model.language_model.layers.3.self_attn.q_norm	256	3,201,024,288
model.language_model.layers.3.self_attn.q_proj	3,145,728	3,204,170,016
model.language_model.layers.3.self_attn.v_proj	393,216	3,204,563,232
model.language_model.layers.4.input_layernorm	1,536	3,204,564,768
model.language_model.layers.4.mlp.down_proj	9,437,184	3,214,001,952
model.language_model.layers.4.mlp.gate_proj	9,437,184	3,223,439,136
model.language_model.layers.4.mlp.up_proj	9,437,184	3,232,876,320
model.language_model.layers.4.per_layer_input_gate	393,216	3,233,269,536
model.language_model.layers.4.per_layer_projection	393,216	3,233,662,752
model.language_model.layers.4. post_attention_layernorm	1,536	3,233,664,288
model.language_model.layers.4. post_feedforward_layernorm	1,536	3,233,665,824
model.language_model.layers.4. post_per_layer_input_norm	1,536	3,233,667,360
model.language_model.layers.4. pre_feedforward_layernorm	1,536	3,233,668,896
model.language_model.layers.4.self_attn.k_norm	512	3,233,669,408
model.language_model.layers.4.self_attn.k_proj	786,432	3,234,455,840
model.language_model.layers.4.self_attn.o_proj	6,291,456	3,240,747,296
model.language_model.layers.4.self_attn.q_norm	512	3,240,747,808
model.language_model.layers.4.self_attn.q_proj	6,291,456	3,247,039,264
model.language_model.layers.4.self_attn.v_proj	786,432	3,247,825,696
model.language_model.layers.5.input_layernorm	1,536	3,247,827,232
model.language_model.layers.5.mlp.down_proj	9,437,184	3,257,264,416
model.language_model.layers.5.mlp.gate_proj	9,437,184	3,266,701,600
model.language_model.layers.5.mlp.up_proj	9,437,184	3,276,138,784

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Module	Params	Running total
model.language_model.layers.5.per_layer_input_gate	393,216	3,276,532,000
model.language_model.layers.5.per_layer_projection	393,216	3,276,925,216
model.language_model.layers.5. post_attention_layernorm	1,536	3,276,926,752
model.language_model.layers.5. post_feedforward_layernorm	1,536	3,276,928,288
model.language_model.layers.5. post_per_layer_input_norm	1,536	3,276,929,824
model.language_model.layers.5. pre_feedforward_layernorm	1,536	3,276,931,360
model.language_model.layers.5.self_attn.k_norm	256	3,276,931,616
model.language_model.layers.5.self_attn.k_proj	393,216	3,277,324,832
model.language_model.layers.5.self_attn.o_proj	3,145,728	3,280,470,560
model.language_model.layers.5.self_attn.q_norm	256	3,280,470,816
model.language_model.layers.5.self_attn.q_proj	3,145,728	3,283,616,544
model.language_model.layers.5.self_attn.v_proj	393,216	3,284,009,760
model.language_model.layers.6.input_layernorm	1,536	3,284,011,296
model.language_model.layers.6.mlp.down_proj	9,437,184	3,293,448,480
model.language_model.layers.6.mlp.gate_proj	9,437,184	3,302,885,664
model.language_model.layers.6.mlp.up_proj	9,437,184	3,312,322,848
model.language_model.layers.6.per_layer_input_gate	393,216	3,312,716,064
model.language_model.layers.6.per_layer_projection	393,216	3,313,109,280
model.language_model.layers.6. post_attention_layernorm	1,536	3,313,110,816
model.language_model.layers.6. post_feedforward_layernorm	1,536	3,313,112,352
model.language_model.layers.6. post_per_layer_input_norm	1,536	3,313,113,888
model.language_model.layers.6. pre_feedforward_layernorm	1,536	3,313,115,424
model.language_model.layers.6.self_attn.k_norm	256	3,313,115,680
model.language_model.layers.6.self_attn.k_proj	393,216	3,313,508,896
model.language_model.layers.6.self_attn.o_proj	3,145,728	3,316,654,624
model.language_model.layers.6.self_attn.q_norm	256	3,316,654,880
model.language_model.layers.6.self_attn.q_proj	3,145,728	3,319,800,608
model.language_model.layers.6.self_attn.v_proj	393,216	3,320,193,824
model.language_model.layers.7.input_layernorm	1,536	3,320,195,360
model.language_model.layers.7.mlp.down_proj	9,437,184	3,329,632,544
model.language_model.layers.7.mlp.gate_proj	9,437,184	3,339,069,728
model.language_model.layers.7.mlp.up_proj	9,437,184	3,348,506,912
model.language_model.layers.7.per_layer_input_gate	393,216	3,348,900,128
model.language_model.layers.7.per_layer_projection	393,216	3,349,293,344
model.language_model.layers.7. post_attention_layernorm	1,536	3,349,294,880
model.language_model.layers.7. post_feedforward_layernorm	1,536	3,349,296,416
model.language_model.layers.7. post_per_layer_input_norm	1,536	3,349,297,952
model.language_model.layers.7. pre_feedforward_layernorm	1,536	3,349,299,488
model.language_model.layers.7.self_attn.k_norm	256	3,349,299,744
model.language_model.layers.7.self_attn.k_proj	393,216	3,349,692,960
model.language_model.layers.7.self_attn.o_proj	3,145,728	3,352,838,688
model.language_model.layers.7.self_attn.q_norm	256	3,352,838,944
model.language_model.layers.7.self_attn.q_proj	3,145,728	3,355,984,672
model.language_model.layers.7.self_attn.v_proj	393,216	3,356,377,888
model.language_model.layers.8.input_layernorm	1,536	3,356,379,424
model.language_model.layers.8.mlp.down_proj	9,437,184	3,365,816,608

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Module	Params	Running total
model.language_model.layers.8.mlp.gate_proj	9,437,184	3,375,253,792
model.language_model.layers.8.mlp.up_proj	9,437,184	3,384,690,976
model.language_model.layers.8.per_layer_input_gate	393,216	3,385,084,192
model.language_model.layers.8.per_layer_projection	393,216	3,385,477,408
model.language_model.layers.8. post_attention_layernorm	1,536	3,385,478,944
model.language_model.layers.8. post_feedforward_layernorm	1,536	3,385,480,480
model.language_model.layers.8. post_per_layer_input_norm	1,536	3,385,482,016
model.language_model.layers.8. pre_feedforward_layernorm	1,536	3,385,483,552
model.language_model.layers.8.self_attn.k_norm	256	3,385,483,808
model.language_model.layers.8.self_attn.k_proj	393,216	3,385,877,024
model.language_model.layers.8.self_attn.o_proj	3,145,728	3,389,022,752
model.language_model.layers.8.self_attn.q_norm	256	3,389,023,008
model.language_model.layers.8.self_attn.q_proj	3,145,728	3,392,168,736
model.language_model.layers.8.self_attn.v_proj	393,216	3,392,561,952
model.language_model.layers.9.input_layernorm	1,536	3,392,563,488
model.language_model.layers.9.mlp.down_proj	9,437,184	3,402,000,672
model.language_model.layers.9.mlp.gate_proj	9,437,184	3,411,437,856
model.language_model.layers.9.mlp.up_proj	9,437,184	3,420,875,040
model.language_model.layers.9.per_layer_input_gate	393,216	3,421,268,256
model.language_model.layers.9.per_layer_projection	393,216	3,421,661,472
model.language_model.layers.9. post_attention_layernorm	1,536	3,421,663,008
model.language_model.layers.9. post_feedforward_layernorm	1,536	3,421,664,544
model.language_model.layers.9. post_per_layer_input_norm	1,536	3,421,666,080
model.language_model.layers.9. pre_feedforward_layernorm	1,536	3,421,667,616
model.language_model.layers.9.self_attn.k_norm	512	3,421,668,128
model.language_model.layers.9.self_attn.k_proj	786,432	3,422,454,560
model.language_model.layers.9.self_attn.o_proj	6,291,456	3,428,746,016
model.language_model.layers.9.self_attn.q_norm	512	3,428,746,528
model.language_model.layers.9.self_attn.q_proj	6,291,456	3,435,037,984
model.language_model.layers.9.self_attn.v_proj	786,432	3,435,824,416
model.language_model.layers.10.input_layernorm	1,536	3,435,825,952
model.language_model.layers.10.mlp.down_proj	9,437,184	3,445,263,136
model.language_model.layers.10.mlp.gate_proj	9,437,184	3,454,700,320
model.language_model.layers.10.mlp.up_proj	9,437,184	3,464,137,504
model.language_model.layers.10.per_layer_input_gate	393,216	3,464,530,720
model.language_model.layers.10.per_layer_projection	393,216	3,464,923,936
model.language_model.layers.10. post_attention_layernorm	1,536	3,464,925,472
model.language_model.layers.10. post_feedforward_layernorm	1,536	3,464,927,008
model.language_model.layers.10. post_per_layer_input_norm	1,536	3,464,928,544
model.language_model.layers.10. pre_feedforward_layernorm	1,536	3,464,930,080
model.language_model.layers.10.self_attn.k_norm	256	3,464,930,336
model.language_model.layers.10.self_attn.k_proj	393,216	3,465,323,552
model.language_model.layers.10.self_attn.o_proj	3,145,728	3,468,469,280
model.language_model.layers.10.self_attn.q_norm	256	3,468,469,536
model.language_model.layers.10.self_attn.q_proj	3,145,728	3,471,615,264
model.language_model.layers.10.self_attn.v_proj	393,216	3,472,008,480

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Module	Params	Running total
model.language_model.layers.11.input_layernorm	1,536	3,472,010,016
model.language_model.layers.11.mlp.down_proj	9,437,184	3,481,447,200
model.language_model.layers.11.mlp.gate_proj	9,437,184	3,490,884,384
model.language_model.layers.11.mlp.up_proj	9,437,184	3,500,321,568
model.language_model.layers.11.per_layer_input_gate	393,216	3,500,714,784
model.language_model.layers.11.per_layer_projection	393,216	3,501,108,000
model.language_model.layers.11.post_attention_layernorm	1,536	3,501,109,536
model.language_model.layers.11.post_feedforward_layernorm	1,536	3,501,111,072
model.language_model.layers.11.post_per_layer_input_norm	1,536	3,501,112,608
model.language_model.layers.11.pre_feedforward_layernorm	1,536	3,501,114,144
model.language_model.layers.11.self_attn.k_norm	256	3,501,114,400
model.language_model.layers.11.self_attn.k_proj	393,216	3,501,507,616
model.language_model.layers.11.self_attn.o_proj	3,145,728	3,504,653,344
model.language_model.layers.11.self_attn.q_norm	256	3,504,653,600
model.language_model.layers.11.self_attn.q_proj	3,145,728	3,507,799,328
model.language_model.layers.11.self_attn.v_proj	393,216	3,508,192,544
model.language_model.layers.12.input_layernorm	1,536	3,508,194,080
model.language_model.layers.12.mlp.down_proj	9,437,184	3,517,631,264
model.language_model.layers.12.mlp.gate_proj	9,437,184	3,527,068,448
model.language_model.layers.12.mlp.up_proj	9,437,184	3,536,505,632
model.language_model.layers.12.per_layer_input_gate	393,216	3,536,898,848
model.language_model.layers.12.per_layer_projection	393,216	3,537,292,064
model.language_model.layers.12.post_attention_layernorm	1,536	3,537,293,600
model.language_model.layers.12.post_feedforward_layernorm	1,536	3,537,295,136
model.language_model.layers.12.post_per_layer_input_norm	1,536	3,537,296,672
model.language_model.layers.12.pre_feedforward_layernorm	1,536	3,537,298,208
model.language_model.layers.12.self_attn.k_norm	256	3,537,298,464
model.language_model.layers.12.self_attn.k_proj	393,216	3,537,691,680
model.language_model.layers.12.self_attn.o_proj	3,145,728	3,540,837,408
model.language_model.layers.12.self_attn.q_norm	256	3,540,837,664
model.language_model.layers.12.self_attn.q_proj	3,145,728	3,543,983,392
model.language_model.layers.12.self_attn.v_proj	393,216	3,544,376,608
model.language_model.layers.13.input_layernorm	1,536	3,544,378,144
model.language_model.layers.13.mlp.down_proj	9,437,184	3,553,815,328
model.language_model.layers.13.mlp.gate_proj	9,437,184	3,563,252,512
model.language_model.layers.13.mlp.up_proj	9,437,184	3,572,689,696
model.language_model.layers.13.per_layer_input_gate	393,216	3,573,082,912
model.language_model.layers.13.per_layer_projection	393,216	3,573,476,128
model.language_model.layers.13.post_attention_layernorm	1,536	3,573,477,664
model.language_model.layers.13.post_feedforward_layernorm	1,536	3,573,479,200
model.language_model.layers.13.post_per_layer_input_norm	1,536	3,573,480,736
model.language_model.layers.13.pre_feedforward_layernorm	1,536	3,573,482,272
model.language_model.layers.13.self_attn.k_norm	256	3,573,482,528
model.language_model.layers.13.self_attn.k_proj	393,216	3,573,875,744
model.language_model.layers.13.self_attn.o_proj	3,145,728	3,577,021,472
model.language_model.layers.13.self_attn.q_norm	256	3,577,021,728

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Module	Params	Running total
model.language_model.layers.13.self_attn.q_proj	3,145,728	3,580,167,456
model.language_model.layers.13.self_attn.v_proj	393,216	3,580,560,672
model.language_model.layers.14.input_layernorm	1,536	3,580,562,208
model.language_model.layers.14.mlp.down_proj	9,437,184	3,589,999,392
model.language_model.layers.14.mlp.gate_proj	9,437,184	3,599,436,576
model.language_model.layers.14.mlp.up_proj	9,437,184	3,608,873,760
model.language_model.layers.14.per_layer_input_gate	393,216	3,609,266,976
model.language_model.layers.14.per_layer_projection	393,216	3,609,660,192
model.language_model.layers.14. post_attention_layernorm	1,536	3,609,661,728
model.language_model.layers.14. post_feedforward_layernorm	1,536	3,609,663,264
model.language_model.layers.14. post_per_layer_input_norm	1,536	3,609,664,800
model.language_model.layers.14. pre_feedforward_layernorm	1,536	3,609,666,336
model.language_model.layers.14.self_attn.k_norm	512	3,609,666,848
model.language_model.layers.14.self_attn.k_proj	786,432	3,610,453,280
model.language_model.layers.14.self_attn.o_proj	6,291,456	3,616,744,736
model.language_model.layers.14.self_attn.q_norm	512	3,616,745,248
model.language_model.layers.14.self_attn.q_proj	6,291,456	3,623,036,704
model.language_model.layers.14.self_attn.v_proj	786,432	3,623,823,136
model.language_model.layers.15.input_layernorm	1,536	3,623,824,672
model.language_model.layers.15.mlp.down_proj	18,874,368	3,642,699,040
model.language_model.layers.15.mlp.gate_proj	18,874,368	3,661,573,408
model.language_model.layers.15.mlp.up_proj	18,874,368	3,680,447,776
model.language_model.layers.15.per_layer_input_gate	393,216	3,680,840,992
model.language_model.layers.15.per_layer_projection	393,216	3,681,234,208
model.language_model.layers.15. post_attention_layernorm	1,536	3,681,235,744
model.language_model.layers.15. post_feedforward_layernorm	1,536	3,681,237,280
model.language_model.layers.15. post_per_layer_input_norm	1,536	3,681,238,816
model.language_model.layers.15. pre_feedforward_layernorm	1,536	3,681,240,352
model.language_model.layers.15.self_attn.o_proj	3,145,728	3,684,386,080
model.language_model.layers.15.self_attn.q_norm	256	3,684,386,336
model.language_model.layers.15.self_attn.q_proj	3,145,728	3,687,532,064
model.language_model.layers.16.input_layernorm	1,536	3,687,533,600
model.language_model.layers.16.mlp.down_proj	18,874,368	3,706,407,968
model.language_model.layers.16.mlp.gate_proj	18,874,368	3,725,282,336
model.language_model.layers.16.mlp.up_proj	18,874,368	3,744,156,704
model.language_model.layers.16.per_layer_input_gate	393,216	3,744,549,920
model.language_model.layers.16.per_layer_projection	393,216	3,744,943,136
model.language_model.layers.16. post_attention_layernorm	1,536	3,744,944,672
model.language_model.layers.16. post_feedforward_layernorm	1,536	3,744,946,208
model.language_model.layers.16. post_per_layer_input_norm	1,536	3,744,947,744
model.language_model.layers.16. pre_feedforward_layernorm	1,536	3,744,949,280
model.language_model.layers.16.self_attn.o_proj	3,145,728	3,748,095,008
model.language_model.layers.16.self_attn.q_norm	256	3,748,095,264
model.language_model.layers.16.self_attn.q_proj	3,145,728	3,751,240,992
model.language_model.layers.17.input_layernorm	1,536	3,751,242,528
model.language_model.layers.17.mlp.down_proj	18,874,368	3,770,116,896

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Module	Params	Running total
model.language_model.layers.17.mlp.gate_proj	18,874,368	3,788,991,264
model.language_model.layers.17.mlp.up_proj	18,874,368	3,807,865,632
model.language_model.layers.17.per_layer_input_gate	393,216	3,808,258,848
model.language_model.layers.17.per_layer_projection	393,216	3,808,652,064
model.language_model.layers.17.post_attention_layernorm	1,536	3,808,653,600
model.language_model.layers.17.post_feedforward_layernorm	1,536	3,808,655,136
model.language_model.layers.17.post_per_layer_input_norm	1,536	3,808,656,672
model.language_model.layers.17.pre_feedforward_layernorm	1,536	3,808,658,208
model.language_model.layers.17.self_attn.o_proj	3,145,728	3,811,803,936
model.language_model.layers.17.self_attn.q_norm	256	3,811,804,192
model.language_model.layers.17.self_attn.q_proj	3,145,728	3,814,949,920
model.language_model.layers.18.input_layernorm	1,536	3,814,951,456
model.language_model.layers.18.mlp.down_proj	18,874,368	3,833,825,824
model.language_model.layers.18.mlp.gate_proj	18,874,368	3,852,700,192
model.language_model.layers.18.mlp.up_proj	18,874,368	3,871,574,560
model.language_model.layers.18.per_layer_input_gate	393,216	3,871,967,776
model.language_model.layers.18.per_layer_projection	393,216	3,872,360,992
model.language_model.layers.18.post_attention_layernorm	1,536	3,872,362,528
model.language_model.layers.18.post_feedforward_layernorm	1,536	3,872,364,064
model.language_model.layers.18.post_per_layer_input_norm	1,536	3,872,365,600
model.language_model.layers.18.pre_feedforward_layernorm	1,536	3,872,367,136
model.language_model.layers.18.self_attn.o_proj	3,145,728	3,875,512,864
model.language_model.layers.18.self_attn.q_norm	256	3,875,513,120
model.language_model.layers.18.self_attn.q_proj	3,145,728	3,878,658,848
model.language_model.layers.19.input_layernorm	1,536	3,878,660,384
model.language_model.layers.19.mlp.down_proj	18,874,368	3,897,534,752
model.language_model.layers.19.mlp.gate_proj	18,874,368	3,916,409,120
model.language_model.layers.19.mlp.up_proj	18,874,368	3,935,283,488
model.language_model.layers.19.per_layer_input_gate	393,216	3,935,676,704
model.language_model.layers.19.per_layer_projection	393,216	3,936,069,920
model.language_model.layers.19.post_attention_layernorm	1,536	3,936,071,456
model.language_model.layers.19.post_feedforward_layernorm	1,536	3,936,072,992
model.language_model.layers.19.post_per_layer_input_norm	1,536	3,936,074,528
model.language_model.layers.19.pre_feedforward_layernorm	1,536	3,936,076,064
model.language_model.layers.19.self_attn.o_proj	6,291,456	3,942,367,520
model.language_model.layers.19.self_attn.q_norm	512	3,942,368,032
model.language_model.layers.19.self_attn.q_proj	6,291,456	3,948,659,488
model.language_model.layers.20.input_layernorm	1,536	3,948,661,024
model.language_model.layers.20.mlp.down_proj	18,874,368	3,967,535,392
model.language_model.layers.20.mlp.gate_proj	18,874,368	3,986,409,760
model.language_model.layers.20.mlp.up_proj	18,874,368	4,005,284,128
model.language_model.layers.20.per_layer_input_gate	393,216	4,005,677,344
model.language_model.layers.20.per_layer_projection	393,216	4,006,070,560
model.language_model.layers.20.post_attention_layernorm	1,536	4,006,072,096

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Module	Params	Running total
model.language_model.layers.20.post_feedforward_layernorm	1,536	4,006,073,632
model.language_model.layers.20.post_per_layer_input_norm	1,536	4,006,075,168
model.language_model.layers.20.pre_feedforward_layernorm	1,536	4,006,076,704
model.language_model.layers.20.self_attn.o_proj	3,145,728	4,009,222,432
model.language_model.layers.20.self_attn.q_norm	256	4,009,222,688
model.language_model.layers.20.self_attn.q_proj	3,145,728	4,012,368,416
model.language_model.layers.21.input_layernorm	1,536	4,012,369,952
model.language_model.layers.21.mlp.down_proj	18,874,368	4,031,244,320
model.language_model.layers.21.mlp.gate_proj	18,874,368	4,050,118,688
model.language_model.layers.21.mlp.up_proj	18,874,368	4,068,993,056
model.language_model.layers.21.per_layer_input_gate	393,216	4,069,386,272
model.language_model.layers.21.per_layer_projection	393,216	4,069,779,488
model.language_model.layers.21.post_attention_layernorm	1,536	4,069,781,024
model.language_model.layers.21.post_feedforward_layernorm	1,536	4,069,782,560
model.language_model.layers.21.post_per_layer_input_norm	1,536	4,069,784,096
model.language_model.layers.21.pre_feedforward_layernorm	1,536	4,069,785,632
model.language_model.layers.21.self_attn.o_proj	3,145,728	4,072,931,360
model.language_model.layers.21.self_attn.q_norm	256	4,072,931,616
model.language_model.layers.21.self_attn.q_proj	3,145,728	4,076,077,344
model.language_model.layers.22.input_layernorm	1,536	4,076,078,880
model.language_model.layers.22.mlp.down_proj	18,874,368	4,094,953,248
model.language_model.layers.22.mlp.gate_proj	18,874,368	4,113,827,616
model.language_model.layers.22.mlp.up_proj	18,874,368	4,132,701,984
model.language_model.layers.22.per_layer_input_gate	393,216	4,133,095,200
model.language_model.layers.22.per_layer_projection	393,216	4,133,488,416
model.language_model.layers.22.post_attention_layernorm	1,536	4,133,489,952
model.language_model.layers.22.post_feedforward_layernorm	1,536	4,133,491,488
model.language_model.layers.22.post_per_layer_input_norm	1,536	4,133,493,024
model.language_model.layers.22.pre_feedforward_layernorm	1,536	4,133,494,560
model.language_model.layers.22.self_attn.o_proj	3,145,728	4,136,640,288
model.language_model.layers.22.self_attn.q_norm	256	4,136,640,544
model.language_model.layers.22.self_attn.q_proj	3,145,728	4,139,786,272
model.language_model.layers.23.input_layernorm	1,536	4,139,787,808
model.language_model.layers.23.mlp.down_proj	18,874,368	4,158,662,176
model.language_model.layers.23.mlp.gate_proj	18,874,368	4,177,536,544
model.language_model.layers.23.mlp.up_proj	18,874,368	4,196,410,912
model.language_model.layers.23.per_layer_input_gate	393,216	4,196,804,128
model.language_model.layers.23.per_layer_projection	393,216	4,197,197,344
model.language_model.layers.23.post_attention_layernorm	1,536	4,197,198,880
model.language_model.layers.23.post_feedforward_layernorm	1,536	4,197,200,416
model.language_model.layers.23.post_per_layer_input_norm	1,536	4,197,201,952
model.language_model.layers.23.pre_feedforward_layernorm	1,536	4,197,203,488
model.language_model.layers.23.self_attn.o_proj	3,145,728	4,200,349,216

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Module	Params	Running total
model.language_model.layers.23.self_attn.q_norm	256	4,200,349,472
model.language_model.layers.23.self_attn.q_proj	3,145,728	4,203,495,200
model.language_model.layers.24.input_layernorm	1,536	4,203,496,736
model.language_model.layers.24.mlp.down_proj	18,874,368	4,222,371,104
model.language_model.layers.24.mlp.gate_proj	18,874,368	4,241,245,472
model.language_model.layers.24.mlp.up_proj	18,874,368	4,260,119,840
model.language_model.layers.24.per_layer_input_gate	393,216	4,260,513,056
model.language_model.layers.24.per_layer_projection	393,216	4,260,906,272
model.language_model.layers.24.post_attention_layernorm	1,536	4,260,907,808
model.language_model.layers.24.post_feedforward_layernorm	1,536	4,260,909,344
model.language_model.layers.24.post_per_layer_input_norm	1,536	4,260,910,880
model.language_model.layers.24.pre_feedforward_layernorm	1,536	4,260,912,416
model.language_model.layers.24.self_attn.o_proj	6,291,456	4,267,203,872
model.language_model.layers.24.self_attn.q_norm	512	4,267,204,384
model.language_model.layers.24.self_attn.q_proj	6,291,456	4,273,495,840
model.language_model.layers.25.input_layernorm	1,536	4,273,497,376
model.language_model.layers.25.mlp.down_proj	18,874,368	4,292,371,744
model.language_model.layers.25.mlp.gate_proj	18,874,368	4,311,246,112
model.language_model.layers.25.mlp.up_proj	18,874,368	4,330,120,480
model.language_model.layers.25.per_layer_input_gate	393,216	4,330,513,696
model.language_model.layers.25.per_layer_projection	393,216	4,330,906,912
model.language_model.layers.25.post_attention_layernorm	1,536	4,330,908,448
model.language_model.layers.25.post_feedforward_layernorm	1,536	4,330,909,984
model.language_model.layers.25.post_per_layer_input_norm	1,536	4,330,911,520
model.language_model.layers.25.pre_feedforward_layernorm	1,536	4,330,913,056
model.language_model.layers.25.self_attn.o_proj	3,145,728	4,334,058,784
model.language_model.layers.25.self_attn.q_norm	256	4,334,059,040
model.language_model.layers.25.self_attn.q_proj	3,145,728	4,337,204,768
model.language_model.layers.26.input_layernorm	1,536	4,337,206,304
model.language_model.layers.26.mlp.down_proj	18,874,368	4,356,080,672
model.language_model.layers.26.mlp.gate_proj	18,874,368	4,374,955,040
model.language_model.layers.26.mlp.up_proj	18,874,368	4,393,829,408
model.language_model.layers.26.per_layer_input_gate	393,216	4,394,222,624
model.language_model.layers.26.per_layer_projection	393,216	4,394,615,840
model.language_model.layers.26.post_attention_layernorm	1,536	4,394,617,376
model.language_model.layers.26.post_feedforward_layernorm	1,536	4,394,618,912
model.language_model.layers.26.post_per_layer_input_norm	1,536	4,394,620,448
model.language_model.layers.26.pre_feedforward_layernorm	1,536	4,394,621,984
model.language_model.layers.26.self_attn.o_proj	3,145,728	4,397,767,712
model.language_model.layers.26.self_attn.q_norm	256	4,397,767,968
model.language_model.layers.26.self_attn.q_proj	3,145,728	4,400,913,696
model.language_model.layers.27.input_layernorm	1,536	4,400,915,232
model.language_model.layers.27.mlp.down_proj	18,874,368	4,419,789,600
model.language_model.layers.27.mlp.gate_proj	18,874,368	4,438,663,968
model.language_model.layers.27.mlp.up_proj	18,874,368	4,457,538,336
model.language_model.layers.27.per_layer_input_gate	393,216	4,457,931,552

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Module	Params	Running total
model.language_model.layers.27.per_layer_projection	393,216	4,458,324,768
model.language_model.layers.27. post_attention_layernorm	1,536	4,458,326,304
model.language_model.layers.27. post_feedforward_layernorm	1,536	4,458,327,840
model.language_model.layers.27. post_per_layer_input_norm	1,536	4,458,329,376
model.language_model.layers.27. pre_feedforward_layernorm	1,536	4,458,330,912
model.language_model.layers.27.self_attn.o_proj	3,145,728	4,461,476,640
model.language_model.layers.27.self_attn.q_norm	256	4,461,476,896
model.language_model.layers.27.self_attn.q_proj	3,145,728	4,464,622,624
model.language_model.layers.28.input_layernorm	1,536	4,464,624,160
model.language_model.layers.28.mlp.down_proj	18,874,368	4,483,498,528
model.language_model.layers.28.mlp.gate_proj	18,874,368	4,502,372,896
model.language_model.layers.28.mlp.up_proj	18,874,368	4,521,247,264
model.language_model.layers.28.per_layer_input_gate	393,216	4,521,640,480
model.language_model.layers.28.per_layer_projection	393,216	4,522,033,696
model.language_model.layers.28. post_attention_layernorm	1,536	4,522,035,232
model.language_model.layers.28. post_feedforward_layernorm	1,536	4,522,036,768
model.language_model.layers.28. post_per_layer_input_norm	1,536	4,522,038,304
model.language_model.layers.28. pre_feedforward_layernorm	1,536	4,522,039,840
model.language_model.layers.28.self_attn.o_proj	3,145,728	4,525,185,568
model.language_model.layers.28.self_attn.q_norm	256	4,525,185,824
model.language_model.layers.28.self_attn.q_proj	3,145,728	4,528,331,552
model.language_model.layers.29.input_layernorm	1,536	4,528,333,088
model.language_model.layers.29.mlp.down_proj	18,874,368	4,547,207,456
model.language_model.layers.29.mlp.gate_proj	18,874,368	4,566,081,824
model.language_model.layers.29.mlp.up_proj	18,874,368	4,584,956,192
model.language_model.layers.29.per_layer_input_gate	393,216	4,585,349,408
model.language_model.layers.29.per_layer_projection	393,216	4,585,742,624
model.language_model.layers.29. post_attention_layernorm	1,536	4,585,744,160
model.language_model.layers.29. post_feedforward_layernorm	1,536	4,585,745,696
model.language_model.layers.29. post_per_layer_input_norm	1,536	4,585,747,232
model.language_model.layers.29. pre_feedforward_layernorm	1,536	4,585,748,768
model.language_model.layers.29.self_attn.o_proj	6,291,456	4,592,040,224
model.language_model.layers.29.self_attn.q_norm	512	4,592,040,736
model.language_model.layers.29.self_attn.q_proj	6,291,456	4,598,332,192
model.language_model.layers.30.input_layernorm	1,536	4,598,333,728
model.language_model.layers.30.mlp.down_proj	18,874,368	4,617,208,096
model.language_model.layers.30.mlp.gate_proj	18,874,368	4,636,082,464
model.language_model.layers.30.mlp.up_proj	18,874,368	4,654,956,832
model.language_model.layers.30.per_layer_input_gate	393,216	4,655,350,048
model.language_model.layers.30.per_layer_projection	393,216	4,655,743,264
model.language_model.layers.30. post_attention_layernorm	1,536	4,655,744,800
model.language_model.layers.30. post_feedforward_layernorm	1,536	4,655,746,336
model.language_model.layers.30. post_per_layer_input_norm	1,536	4,655,747,872

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Module	Params	Running total
model.language_model.layers.30. pre_feedforward_layernorm	1,536	4,655,749,408
model.language_model.layers.30.self_attn.o_proj	3,145,728	4,658,895,136
model.language_model.layers.30.self_attn.q_norm	256	4,658,895,392
model.language_model.layers.30.self_attn.q_proj	3,145,728	4,662,041,120
model.language_model.layers.31.input_layernorm	1,536	4,662,042,656
model.language_model.layers.31.mlp.down_proj	18,874,368	4,680,917,024
model.language_model.layers.31.mlp.gate_proj	18,874,368	4,699,791,392
model.language_model.layers.31.mlp.up_proj	18,874,368	4,718,665,760
model.language_model.layers.31.per_layer_input_gate	393,216	4,719,058,976
model.language_model.layers.31.per_layer_projection	393,216	4,719,452,192
model.language_model.layers.31. post_attention_layernorm	1,536	4,719,453,728
model.language_model.layers.31. post_feedforward_layernorm	1,536	4,719,455,264
model.language_model.layers.31. post_per_layer_input_norm	1,536	4,719,456,800
model.language_model.layers.31. pre_feedforward_layernorm	1,536	4,719,458,336
model.language_model.layers.31.self_attn.o_proj	3,145,728	4,722,604,064
model.language_model.layers.31.self_attn.q_norm	256	4,722,604,320
model.language_model.layers.31.self_attn.q_proj	3,145,728	4,725,750,048
model.language_model.layers.32.input_layernorm	1,536	4,725,751,584
model.language_model.layers.32.mlp.down_proj	18,874,368	4,744,625,952
model.language_model.layers.32.mlp.gate_proj	18,874,368	4,763,500,320
model.language_model.layers.32.mlp.up_proj	18,874,368	4,782,374,688
model.language_model.layers.32.per_layer_input_gate	393,216	4,782,767,904
model.language_model.layers.32.per_layer_projection	393,216	4,783,161,120
model.language_model.layers.32. post_attention_layernorm	1,536	4,783,162,656
model.language_model.layers.32. post_feedforward_layernorm	1,536	4,783,164,192
model.language_model.layers.32. post_per_layer_input_norm	1,536	4,783,165,728
model.language_model.layers.32. pre_feedforward_layernorm	1,536	4,783,167,264
model.language_model.layers.32.self_attn.o_proj	3,145,728	4,786,312,992
model.language_model.layers.32.self_attn.q_norm	256	4,786,313,248
model.language_model.layers.32.self_attn.q_proj	3,145,728	4,789,458,976
model.language_model.layers.33.input_layernorm	1,536	4,789,460,512
model.language_model.layers.33.mlp.down_proj	18,874,368	4,808,334,880
model.language_model.layers.33.mlp.gate_proj	18,874,368	4,827,209,248
model.language_model.layers.33.mlp.up_proj	18,874,368	4,846,083,616
model.language_model.layers.33.per_layer_input_gate	393,216	4,846,476,832
model.language_model.layers.33.per_layer_projection	393,216	4,846,870,048
model.language_model.layers.33. post_attention_layernorm	1,536	4,846,871,584
model.language_model.layers.33. post_feedforward_layernorm	1,536	4,846,873,120
model.language_model.layers.33. post_per_layer_input_norm	1,536	4,846,874,656
model.language_model.layers.33. pre_feedforward_layernorm	1,536	4,846,876,192
model.language_model.layers.33.self_attn.o_proj	3,145,728	4,850,021,920
model.language_model.layers.33.self_attn.q_norm	256	4,850,022,176
model.language_model.layers.33.self_attn.q_proj	3,145,728	4,853,167,904
model.language_model.layers.34.input_layernorm	1,536	4,853,169,440
model.language_model.layers.34.mlp.down_proj	18,874,368	4,872,043,808

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Module	Params	Running total
model.language_model.layers.34.mlp.gate_proj	18,874,368	4,890,918,176
model.language_model.layers.34.mlp.up_proj	18,874,368	4,909,792,544
model.language_model.layers.34.per_layer_input_gate	393,216	4,910,185,760
model.language_model.layers.34.per_layer_projection	393,216	4,910,578,976
model.language_model.layers.34. post_attention_layernorm	1,536	4,910,580,512
model.language_model.layers.34. post_feedforward_layernorm	1,536	4,910,582,048
model.language_model.layers.34. post_per_layer_input_norm	1,536	4,910,583,584
model.language_model.layers.34. pre_feedforward_layernorm	1,536	4,910,585,120
model.language_model.layers.34.self_attn.o_proj	6,291,456	4,916,876,576
model.language_model.layers.34.self_attn.q_norm	512	4,916,877,088
model.language_model.layers.34.self_attn.q_proj	6,291,456	4,923,168,544
model.language_model.norm	1,536	4,923,170,080
model.language_model.per_layer_model_projection	13,762,560	4,936,932,640
model.language_model.per_layer_projection_norm	256	4,936,932,896
model.vision_tower.encoder.layers.0.input_layernorm	768	4,936,933,664
model.vision_tower.encoder.layers.0.mlp.down_proj. linear	2,359,296	4,939,292,960
model.vision_tower.encoder.layers.0.mlp.gate_proj. linear	2,359,296	4,941,652,256
model.vision_tower.encoder.layers.0.mlp.up_proj. linear	2,359,296	4,944,011,552
model.vision_tower.encoder.layers.0. post_attention_layernorm	768	4,944,012,320
model.vision_tower.encoder.layers.0. post_feedforward_layernorm	768	4,944,013,088
model.vision_tower.encoder.layers.0. pre_feedforward_layernorm	768	4,944,013,856
model.vision_tower.encoder.layers.0.self_attn. k_norm	64	4,944,013,920
model.vision_tower.encoder.layers.0.self_attn. k_proj.linear	589,824	4,944,603,744
model.vision_tower.encoder.layers.0.self_attn. o_proj.linear	589,824	4,945,193,568
model.vision_tower.encoder.layers.0.self_attn. q_norm	64	4,945,193,632
model.vision_tower.encoder.layers.0.self_attn. q_proj.linear	589,824	4,945,783,456
model.vision_tower.encoder.layers.0.self_attn. v_proj.linear	589,824	4,946,373,280
model.vision_tower.encoder.layers.1.input_layernorm	768	4,946,374,048
model.vision_tower.encoder.layers.1.mlp.down_proj. linear	2,359,296	4,948,733,344
model.vision_tower.encoder.layers.1.mlp.gate_proj. linear	2,359,296	4,951,092,640
model.vision_tower.encoder.layers.1.mlp.up_proj. linear	2,359,296	4,953,451,936
model.vision_tower.encoder.layers.1. post_attention_layernorm	768	4,953,452,704
model.vision_tower.encoder.layers.1. post_feedforward_layernorm	768	4,953,453,472
model.vision_tower.encoder.layers.1. pre_feedforward_layernorm	768	4,953,454,240
model.vision_tower.encoder.layers.1.self_attn. k_norm	64	4,953,454,304

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Module	Params	Running total
model.vision_tower.encoder.layers.1.self_attn.k_proj.linear	589,824	4,954,044,128
model.vision_tower.encoder.layers.1.self_attn.o_proj.linear	589,824	4,954,633,952
model.vision_tower.encoder.layers.1.self_attn.q_norm	64	4,954,634,016
model.vision_tower.encoder.layers.1.self_attn.q_proj.linear	589,824	4,955,223,840
model.vision_tower.encoder.layers.1.self_attn.v_proj.linear	589,824	4,955,813,664
model.vision_tower.encoder.layers.2.input_layernorm	768	4,955,814,432
model.vision_tower.encoder.layers.2.mlp.down_proj.linear	2,359,296	4,958,173,728
model.vision_tower.encoder.layers.2.mlp.gate_proj.linear	2,359,296	4,960,533,024
model.vision_tower.encoder.layers.2.mlp.up_proj.linear	2,359,296	4,962,892,320
model.vision_tower.encoder.layers.2.post_attention_layernorm	768	4,962,893,088
model.vision_tower.encoder.layers.2.post_feedforward_layernorm	768	4,962,893,856
model.vision_tower.encoder.layers.2.pre_feedforward_layernorm	768	4,962,894,624
model.vision_tower.encoder.layers.2.self_attn.k_norm	64	4,962,894,688
model.vision_tower.encoder.layers.2.self_attn.k_proj.linear	589,824	4,963,484,512
model.vision_tower.encoder.layers.2.self_attn.o_proj.linear	589,824	4,964,074,336
model.vision_tower.encoder.layers.2.self_attn.q_norm	64	4,964,074,400
model.vision_tower.encoder.layers.2.self_attn.q_proj.linear	589,824	4,964,664,224
model.vision_tower.encoder.layers.2.self_attn.v_proj.linear	589,824	4,965,254,048
model.vision_tower.encoder.layers.3.input_layernorm	768	4,965,254,816
model.vision_tower.encoder.layers.3.mlp.down_proj.linear	2,359,296	4,967,614,112
model.vision_tower.encoder.layers.3.mlp.gate_proj.linear	2,359,296	4,969,973,408
model.vision_tower.encoder.layers.3.mlp.up_proj.linear	2,359,296	4,972,332,704
model.vision_tower.encoder.layers.3.post_attention_layernorm	768	4,972,333,472
model.vision_tower.encoder.layers.3.post_feedforward_layernorm	768	4,972,334,240
model.vision_tower.encoder.layers.3.pre_feedforward_layernorm	768	4,972,335,008
model.vision_tower.encoder.layers.3.self_attn.k_norm	64	4,972,335,072
model.vision_tower.encoder.layers.3.self_attn.k_proj.linear	589,824	4,972,924,896
model.vision_tower.encoder.layers.3.self_attn.o_proj.linear	589,824	4,973,514,720
model.vision_tower.encoder.layers.3.self_attn.q_norm	64	4,973,514,784
model.vision_tower.encoder.layers.3.self_attn.q_proj.linear	589,824	4,974,104,608

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Module	Params	Running total
model.vision_tower.encoder.layers.3.self_attn.v_proj.linear	589,824	4,974,694,432
model.vision_tower.encoder.layers.4.input_layernorm	768	4,974,695,200
model.vision_tower.encoder.layers.4.mlp.down_proj.linear	2,359,296	4,977,054,496
model.vision_tower.encoder.layers.4.mlp.gate_proj.linear	2,359,296	4,979,413,792
model.vision_tower.encoder.layers.4.mlp.up_proj.linear	2,359,296	4,981,773,088
model.vision_tower.encoder.layers.4.post_attention_layernorm	768	4,981,773,856
model.vision_tower.encoder.layers.4.post_feedforward_layernorm	768	4,981,774,624
model.vision_tower.encoder.layers.4.pre_feedforward_layernorm	768	4,981,775,392
model.vision_tower.encoder.layers.4.self_attn.k_norm	64	4,981,775,456
model.vision_tower.encoder.layers.4.self_attn.k_proj.linear	589,824	4,982,365,280
model.vision_tower.encoder.layers.4.self_attn.o_proj.linear	589,824	4,982,955,104
model.vision_tower.encoder.layers.4.self_attn.q_norm	64	4,982,955,168
model.vision_tower.encoder.layers.4.self_attn.q_proj.linear	589,824	4,983,544,992
model.vision_tower.encoder.layers.4.self_attn.v_proj.linear	589,824	4,984,134,816
model.vision_tower.encoder.layers.5.input_layernorm	768	4,984,135,584
model.vision_tower.encoder.layers.5.mlp.down_proj.linear	2,359,296	4,986,494,880
model.vision_tower.encoder.layers.5.mlp.gate_proj.linear	2,359,296	4,988,854,176
model.vision_tower.encoder.layers.5.mlp.up_proj.linear	2,359,296	4,991,213,472
model.vision_tower.encoder.layers.5.post_attention_layernorm	768	4,991,214,240
model.vision_tower.encoder.layers.5.post_feedforward_layernorm	768	4,991,215,008
model.vision_tower.encoder.layers.5.pre_feedforward_layernorm	768	4,991,215,776
model.vision_tower.encoder.layers.5.self_attn.k_norm	64	4,991,215,840
model.vision_tower.encoder.layers.5.self_attn.k_proj.linear	589,824	4,991,805,664
model.vision_tower.encoder.layers.5.self_attn.o_proj.linear	589,824	4,992,395,488
model.vision_tower.encoder.layers.5.self_attn.q_norm	64	4,992,395,552
model.vision_tower.encoder.layers.5.self_attn.q_proj.linear	589,824	4,992,985,376
model.vision_tower.encoder.layers.5.self_attn.v_proj.linear	589,824	4,993,575,200
model.vision_tower.encoder.layers.6.input_layernorm	768	4,993,575,968
model.vision_tower.encoder.layers.6.mlp.down_proj.linear	2,359,296	4,995,935,264
model.vision_tower.encoder.layers.6.mlp.gate_proj.linear	2,359,296	4,998,294,560

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Module	Params	Running total
model.vision_tower.encoder.layers.6.mlp.up_proj.linear	2,359,296	5,000,653,856
model.vision_tower.encoder.layers.6.post_attention_layernorm	768	5,000,654,624
model.vision_tower.encoder.layers.6.post_feedforward_layernorm	768	5,000,655,392
model.vision_tower.encoder.layers.6.pre_feedforward_layernorm	768	5,000,656,160
model.vision_tower.encoder.layers.6.self_attn.k_norm	64	5,000,656,224
model.vision_tower.encoder.layers.6.self_attn.k_proj.linear	589,824	5,001,246,048
model.vision_tower.encoder.layers.6.self_attn.o_proj.linear	589,824	5,001,835,872
model.vision_tower.encoder.layers.6.self_attn.q_norm	64	5,001,835,936
model.vision_tower.encoder.layers.6.self_attn.q_proj.linear	589,824	5,002,425,760
model.vision_tower.encoder.layers.6.self_attn.v_proj.linear	589,824	5,003,015,584
model.vision_tower.encoder.layers.7.input_layernorm	768	5,003,016,352
model.vision_tower.encoder.layers.7.mlp.down_proj.linear	2,359,296	5,005,375,648
model.vision_tower.encoder.layers.7.mlp.gate_proj.linear	2,359,296	5,007,734,944
model.vision_tower.encoder.layers.7.mlp.up_proj.linear	2,359,296	5,010,094,240
model.vision_tower.encoder.layers.7.post_attention_layernorm	768	5,010,095,008
model.vision_tower.encoder.layers.7.post_feedforward_layernorm	768	5,010,095,776
model.vision_tower.encoder.layers.7.pre_feedforward_layernorm	768	5,010,096,544
model.vision_tower.encoder.layers.7.self_attn.k_norm	64	5,010,096,608
model.vision_tower.encoder.layers.7.self_attn.k_proj.linear	589,824	5,010,686,432
model.vision_tower.encoder.layers.7.self_attn.o_proj.linear	589,824	5,011,276,256
model.vision_tower.encoder.layers.7.self_attn.q_norm	64	5,011,276,320
model.vision_tower.encoder.layers.7.self_attn.q_proj.linear	589,824	5,011,866,144
model.vision_tower.encoder.layers.7.self_attn.v_proj.linear	589,824	5,012,455,968
model.vision_tower.encoder.layers.8.input_layernorm	768	5,012,456,736
model.vision_tower.encoder.layers.8.mlp.down_proj.linear	2,359,296	5,014,816,032
model.vision_tower.encoder.layers.8.mlp.gate_proj.linear	2,359,296	5,017,175,328
model.vision_tower.encoder.layers.8.mlp.up_proj.linear	2,359,296	5,019,534,624
model.vision_tower.encoder.layers.8.post_attention_layernorm	768	5,019,535,392
model.vision_tower.encoder.layers.8.post_feedforward_layernorm	768	5,019,536,160
model.vision_tower.encoder.layers.8.pre_feedforward_layernorm	768	5,019,536,928

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Module	Params	Running total
model.vision_tower.encoder.layers.8.self_attn.k_norm	64	5,019,536,992
model.vision_tower.encoder.layers.8.self_attn.k_proj.linear	589,824	5,020,126,816
model.vision_tower.encoder.layers.8.self_attn.o_proj.linear	589,824	5,020,716,640
model.vision_tower.encoder.layers.8.self_attn.q_norm	64	5,020,716,704
model.vision_tower.encoder.layers.8.self_attn.q_proj.linear	589,824	5,021,306,528
model.vision_tower.encoder.layers.8.self_attn.v_proj.linear	589,824	5,021,896,352
model.vision_tower.encoder.layers.9.input_layernorm	768	5,021,897,120
model.vision_tower.encoder.layers.9.mlp.down_proj.linear	2,359,296	5,024,256,416
model.vision_tower.encoder.layers.9.mlp.gate_proj.linear	2,359,296	5,026,615,712
model.vision_tower.encoder.layers.9.mlp.up_proj.linear	2,359,296	5,028,975,008
model.vision_tower.encoder.layers.9.post_attention_layernorm	768	5,028,975,776
model.vision_tower.encoder.layers.9.post_feedforward_layernorm	768	5,028,976,544
model.vision_tower.encoder.layers.9.pre_feedforward_layernorm	768	5,028,977,312
model.vision_tower.encoder.layers.9.self_attn.k_norm	64	5,028,977,376
model.vision_tower.encoder.layers.9.self_attn.k_proj.linear	589,824	5,029,567,200
model.vision_tower.encoder.layers.9.self_attn.o_proj.linear	589,824	5,030,157,024
model.vision_tower.encoder.layers.9.self_attn.q_norm	64	5,030,157,088
model.vision_tower.encoder.layers.9.self_attn.q_proj.linear	589,824	5,030,746,912
model.vision_tower.encoder.layers.9.self_attn.v_proj.linear	589,824	5,031,336,736
model.vision_tower.encoder.layers.10.input_layernorm	768	5,031,337,504
model.vision_tower.encoder.layers.10.mlp.down_proj.linear	2,359,296	5,033,696,800
model.vision_tower.encoder.layers.10.mlp.gate_proj.linear	2,359,296	5,036,056,096
model.vision_tower.encoder.layers.10.mlp.up_proj.linear	2,359,296	5,038,415,392
model.vision_tower.encoder.layers.10.post_attention_layernorm	768	5,038,416,160
model.vision_tower.encoder.layers.10.post_feedforward_layernorm	768	5,038,416,928
model.vision_tower.encoder.layers.10.pre_feedforward_layernorm	768	5,038,417,696
model.vision_tower.encoder.layers.10.self_attn.k_norm	64	5,038,417,760
model.vision_tower.encoder.layers.10.self_attn.k_proj.linear	589,824	5,039,007,584
model.vision_tower.encoder.layers.10.self_attn.o_proj.linear	589,824	5,039,597,408

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Module	Params	Running total
model.vision_tower.encoder.layers.10.self_attn.q_norm	64	5,039,597,472
model.vision_tower.encoder.layers.10.self_attn.q_proj.linear	589,824	5,040,187,296
model.vision_tower.encoder.layers.10.self_attn.v_proj.linear	589,824	5,040,777,120
model.vision_tower.encoder.layers.11.input_layernorm	768	5,040,777,888
model.vision_tower.encoder.layers.11.mlp.down_proj.linear	2,359,296	5,043,137,184
model.vision_tower.encoder.layers.11.mlp.gate_proj.linear	2,359,296	5,045,496,480
model.vision_tower.encoder.layers.11.mlp.up_proj.linear	2,359,296	5,047,855,776
model.vision_tower.encoder.layers.11.post_attention_layernorm	768	5,047,856,544
model.vision_tower.encoder.layers.11.post_feedforward_layernorm	768	5,047,857,312
model.vision_tower.encoder.layers.11.pre_feedforward_layernorm	768	5,047,858,080
model.vision_tower.encoder.layers.11.self_attn.k_norm	64	5,047,858,144
model.vision_tower.encoder.layers.11.self_attn.k_proj.linear	589,824	5,048,447,968
model.vision_tower.encoder.layers.11.self_attn.o_proj.linear	589,824	5,049,037,792
model.vision_tower.encoder.layers.11.self_attn.q_norm	64	5,049,037,856
model.vision_tower.encoder.layers.11.self_attn.q_proj.linear	589,824	5,049,627,680
model.vision_tower.encoder.layers.11.self_attn.v_proj.linear	589,824	5,050,217,504
model.vision_tower.encoder.layers.12.input_layernorm	768	5,050,218,272
model.vision_tower.encoder.layers.12.mlp.down_proj.linear	2,359,296	5,052,577,568
model.vision_tower.encoder.layers.12.mlp.gate_proj.linear	2,359,296	5,054,936,864
model.vision_tower.encoder.layers.12.mlp.up_proj.linear	2,359,296	5,057,296,160
model.vision_tower.encoder.layers.12.post_attention_layernorm	768	5,057,296,928
model.vision_tower.encoder.layers.12.post_feedforward_layernorm	768	5,057,297,696
model.vision_tower.encoder.layers.12.pre_feedforward_layernorm	768	5,057,298,464
model.vision_tower.encoder.layers.12.self_attn.k_norm	64	5,057,298,528
model.vision_tower.encoder.layers.12.self_attn.k_proj.linear	589,824	5,057,888,352
model.vision_tower.encoder.layers.12.self_attn.o_proj.linear	589,824	5,058,478,176
model.vision_tower.encoder.layers.12.self_attn.q_norm	64	5,058,478,240
model.vision_tower.encoder.layers.12.self_attn.q_proj.linear	589,824	5,059,068,064
model.vision_tower.encoder.layers.12.self_attn.v_proj.linear	589,824	5,059,657,888

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Module	Params	Running total
model.vision_tower.encoder.layers.13.input_layernorm	768	5,059,658,656
model.vision_tower.encoder.layers.13.mlp.down_proj.linear	2,359,296	5,062,017,952
model.vision_tower.encoder.layers.13.mlp.gate_proj.linear	2,359,296	5,064,377,248
model.vision_tower.encoder.layers.13.mlp.up_proj.linear	2,359,296	5,066,736,544
model.vision_tower.encoder.layers.13.post_attention_layernorm	768	5,066,737,312
model.vision_tower.encoder.layers.13.post_feedforward_layernorm	768	5,066,738,080
model.vision_tower.encoder.layers.13.pre_feedforward_layernorm	768	5,066,738,848
model.vision_tower.encoder.layers.13.self_attn.k_norm	64	5,066,738,912
model.vision_tower.encoder.layers.13.self_attn.k_proj.linear	589,824	5,067,328,736
model.vision_tower.encoder.layers.13.self_attn.o_proj.linear	589,824	5,067,918,560
model.vision_tower.encoder.layers.13.self_attn.q_norm	64	5,067,918,624
model.vision_tower.encoder.layers.13.self_attn.q_proj.linear	589,824	5,068,508,448
model.vision_tower.encoder.layers.13.self_attn.v_proj.linear	589,824	5,069,098,272
model.vision_tower.encoder.layers.14.input_layernorm	768	5,069,099,040
model.vision_tower.encoder.layers.14.mlp.down_proj.linear	2,359,296	5,071,458,336
model.vision_tower.encoder.layers.14.mlp.gate_proj.linear	2,359,296	5,073,817,632
model.vision_tower.encoder.layers.14.mlp.up_proj.linear	2,359,296	5,076,176,928
model.vision_tower.encoder.layers.14.post_attention_layernorm	768	5,076,177,696
model.vision_tower.encoder.layers.14.post_feedforward_layernorm	768	5,076,178,464
model.vision_tower.encoder.layers.14.pre_feedforward_layernorm	768	5,076,179,232
model.vision_tower.encoder.layers.14.self_attn.k_norm	64	5,076,179,296
model.vision_tower.encoder.layers.14.self_attn.k_proj.linear	589,824	5,076,769,120
model.vision_tower.encoder.layers.14.self_attn.o_proj.linear	589,824	5,077,358,944
model.vision_tower.encoder.layers.14.self_attn.q_norm	64	5,077,359,008
model.vision_tower.encoder.layers.14.self_attn.q_proj.linear	589,824	5,077,948,832
model.vision_tower.encoder.layers.14.self_attn.v_proj.linear	589,824	5,078,538,656
model.vision_tower.encoder.layers.15.input_layernorm	768	5,078,539,424
model.vision_tower.encoder.layers.15.mlp.down_proj.linear	2,359,296	5,080,898,720
model.vision_tower.encoder.layers.15.mlp.gate_proj.linear	2,359,296	5,083,258,016

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Module	Params	Running total
model.vision_tower.encoder.layers.15.mlp.up_proj.linear	2,359,296	5,085,617,312
model.vision_tower.encoder.layers.15.post_attention_layernorm	768	5,085,618,080
model.vision_tower.encoder.layers.15.post_feedforward_layernorm	768	5,085,618,848
model.vision_tower.encoder.layers.15.pre_feedforward_layernorm	768	5,085,619,616
model.vision_tower.encoder.layers.15.self_attn.k_norm	64	5,085,619,680
model.vision_tower.encoder.layers.15.self_attn.k_proj.linear	589,824	5,086,209,504
model.vision_tower.encoder.layers.15.self_attn.o_proj.linear	589,824	5,086,799,328
model.vision_tower.encoder.layers.15.self_attn.q_norm	64	5,086,799,392
model.vision_tower.encoder.layers.15.self_attn.q_proj.linear	589,824	5,087,389,216
model.vision_tower.encoder.layers.15.self_attn.v_proj.linear	589,824	5,087,979,040
model.vision_tower.patch_embedder.input_proj	589,824	5,088,568,864
model.vision_tower.patch_embedder	15,728,640	5,104,297,504